What can composition and transport explain about scaling laws?

Theory · Framework paper

Recursive Dynamics · The ARC Theory · Law I, the ARC Principle, a law under test

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Abstract

Predicting how a system scales can help us decide whether resources and correction can keep pace with its growth. This framework paper asks which composition and transport assumptions justify a scaling prediction before the response is observed. Four classical continuous homomorphism problems constrain functional families; they do not classify every physical law. A global logarithmic-defect stability result and a regular-variation limit connect exact equations to specified approximate and asymptotic cases. For a declared steady-flow network, transport accounting gives $C = B\bar{\ell}_q$. Demand scaling $B \sim L^{\mu}$ and mean-path scaling $\bar{\ell}_q \sim L^{\zeta}$ then give an asymptotic throughput-cost exponent $\mu/(\mu+\zeta)$. The familiar $d/(d+1)$ requires further assumptions, and inventory and body mass require separate bridges. Biological measurements are compared on their own variables; pooled fungal confirmation is withdrawn. The Friedmann reparametrisation remains an identity, with acceleration distinguished from exponential expansion. Fixed rules can generate superlinear trajectories, so neither physical dimension nor an architectural label fixes an AI capability exponent. The contribution is a conditional synthesis and a prospective transport test with independently measured premises, held-out predictions and serious rivals. Existing prediction histories, corrections and qualified companion results are retained. No universal physical ceiling, alignment guarantee or new empirical validation is established.

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1 Introduction

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. People should retain oversight, containment and shutdown for as long as they physically can. The research question here is one part of that programme: can information about how gains combine predict later scaling, before a larger system is built or relied on?

A practical example is a supply network. Increasing its size can increase both the demand served and the distance resources travel. Predicting throughput from inventory requires knowing which of those quantities changes, how routes are chosen and whether transport speed changes. A three-dimensional embedding alone does not answer those questions. The same discipline applies to an AI experiment: capability, recursive depth, inference cost and correction burden are distinct observables.

This paper is a theoretical and methodological synthesis across mathematics, transport, biology and computation. It distinguishes three steps: an equation constrains a family of functions; a mechanism supplies the assumptions of that equation; and measurements test whether the mechanism predicts observations. The first step can be a theorem while the later steps remain unresolved. A repeated numerical exponent alone does not join them.

The contributions are:

- A precise account of four classical composition identities, their domains and regularity, with scoped stability and asymptotic extensions.
- An explicit transport calculation showing which demand and path assumptions yield $d/(d+1)$, and which extra assumptions connect transport cost to inventory or body mass.
- Source-specific biological and physical comparisons that separate measurements from coordinate identities and from historical unsupported inferences.
- One prospective test in which independently measured premises predict held-out transport behaviour. The earlier numbered predictions and the full neural-scaling withdrawal remain part of the record.

The mathematics supplies conditional results, not a universal physical classification or a numerical safety guarantee. Sections 4 and 6 establish the formal tools; Sections 7 to 11 examine their scope through examples. Sections 12 and 14 state the evidence and the next test. The appendices preserve the detailed provenance, programme placement and superseded argument.

2 Related work and the contribution

A scaling explanation must do more than reproduce a familiar fraction. It must identify the quantities being related, the rule they obey, the assumptions that select an exponent and the measurement that could show the explanation is wrong. Within Recursive Dynamics, this paper develops those questions for Law I, the ARC Principle, a law under test. It connects three levels of argument: a composition equation that constrains a functional form, a specified transport model that supplies an exponent, and a prospective test of whether that model applies. Earlier work establishes much of the mathematics and several of the domain models. Their differences are part of the explanation.

2.1 Earlier work

Composition and scaling limits. The continuous Cauchy equations give the four homomorphism problems stated in Section 4 (Cauchy 1821). The chosen input and output operations, their domains and regularity conditions do the work. In the multiplicative case the result is a power-law family; it does not choose the exponent. The connection between rescaling and power laws is established mathematics:

Newman (2005, section III.E) discusses it for scale-free distributions, whose inference problem is distinct from the regression of one physical quantity against another.

Approximate composition and asymptotic scaling also have established theories. Hyers (1941) gives stability for an additive equation under a global defect bound. Section 4 states and proves the real-valued consequence needed here, with its logarithmic error metric and its limits for finite observations. Regular variation describes positive functions whose scale ratios converge to a power, allowing a slowly varying factor at finite scale (Bingham et al. 1987). These are mathematical results with premises to check, not evidence that every observed scaling relationship meets them.

Ostaszewski (2016) explicitly connects functional equations, homomorphisms, generalised group operations and regular variation. The author's version 2 preprint is dated 7 November 2014; the article was published online on 20 June 2015 and in the 2016 journal volume. This prior connection is part of the foundation of the present paper. The comparison here uses the scope stated in that work's opening discussion; it does not import its later technical theorems as results about physical or computational systems.

Branching and transport models. The expression $d/(d + 1)$ is earlier work's. West, Brown and Enquist (1997) derive three-quarter scaling for metabolic rates from transport through space-filling fractal networks of branching tubes, with energy dissipation minimised. They state the generalisation to $D/(D + 1)$ when the network fills a D-dimensional volume. West, Brown and Enquist (1999) obtain the same form in D dimensions from a more general treatment of hierarchical-network geometry rather than hydrodynamics. The models have stated limits: the 1997 treatment predicts small deviations in the smallest mammals, and the 1999 treatment does not expect its relationships to apply to filamentous algae and fungi. A fungal application therefore needs its own premises.

Banavar, Maritan and Rinaldo (1999) derive a relationship between size and flow in networks with local connectivity. In the most efficient class they consider, metabolic rate scales as mass to the power $D/(D + 1)$. Their bound concerns the nutrient-delivery rate and the volume of blood in the network; it is not a bound on every observable of a physical system. They also compare their relation with river-drainage observations in $D = 2$ and identify other distributed transport systems to which their reasoning may apply. Banavar and co-authors (2002) develop the resource-distribution account further and explicitly work through the $D = 1$ case. Neither that case nor the fractions $1/2$, $2/3$ and $3/4$ are new predictions of the present paper.

The velocity assumption matters even within a three-dimensional supply model. In the two models of Banavar et al. (2010), constant flow velocity across body sizes gives a metabolic exponent of $2/3$; allowing velocity to scale with mass to the power $1/12$, its maximum in those models, gives $3/4$. Their geometric, steady-supply and transport assumptions remain part of the result. Three-dimensional embedding alone therefore does not select one metabolic exponent. Section 6 keeps flow-weighted transport cost, inventory in transit and body mass separate so that the corresponding assumptions can be seen.

Other mechanisms and other meanings of dimension. Demetrius (2003, 2006) and Demetrius and Tuszynski (2010) develop a quantum-statistical account using coupled energy-transducing oscillators and the Debye model. The 2010 paper derives $d/(d + 1)$ in the regime it calls quantum, with d the dimension of the physical space in which its oscillator network is embedded, and states the values $1/2$, $2/3$ and $3/4$ for $d = 1$, 2 and 3. Its classical regime is isometric. These are different physical assumptions from vascular branching. The flow-accounting calculation in Section 6 does not establish their equivalence or turn their shared expression into separate empirical confirmations. Agutter and Tuszynski (2011) review the competing explanations.

In Bettencourt (2013), urban land area scales with population to the power $D/(D + H)$, where D

is the embedding dimension and H is the fractal dimension of travel paths. The case $D = 2$, $H = 1$ gives $2/3$; the expression becomes $D/(D + 1)$ only when $H = 1$. The same framework gives superlinear socioeconomic rates, with exponent $7/6$ in that case. Land area, socioeconomic output and metabolic rate are distinct observables. Their exponents cannot be exchanged without the corresponding model and measurement definitions.

Zhao (2022) derives an $n/(n + 1)$ scaling law for an n-dimensional system from producer-consumer dynamics. That account places biomass production, eukaryote growth, mammalian brain size and the distribution of public facilities in cities within one proposed universality class, and places linear and superlinear scaling outside that class. Its domain statement belongs to that model; it does not establish that every hierarchical network falls within it.

He and Zhang (2004) derive a $4/5$ law for human-brain allometry from the fractal dimension of a brain cell's contour. This is a different geometrical quantity from the transport-network dimension in Section 6. An equal algebraic form does not identify the two quantities or establish the same mechanism.

Finite size and empirical limits. Maino et al. (2014) derive, from Dynamic Energy Budget theory, an equation relating metabolic rate to mass that is the same as the finite-size form of the West, Brown and Enquist model. It approaches the three-quarter power only at large mass. They stress that metabolic rate is not a power function of mass under either framework. The distinction between a finite-range curve and an asymptotic exponent is therefore substantive, not a plotting detail.

The West, Brown and Enquist derivation has been questioned (Dodds et al. 2001; Kozłowski and Konarzewski 2004), and Savage et al. (2008) show that its three-quarter exponent holds only in the limit of infinite network size. Empirical comparisons also require the competing evidence: White and Seymour (2003) report mammalian basal-metabolic estimates compatible with $2/3$ under their stated adjustments and exclusions, while Kolokotrones et al. (2010) find curvature on logarithmic axes. Section 12 sets out the relevant measurements and their limits. A shared fraction across models is not a substitute for choosing between their premises against evidence.

2.2 What this paper adds

The contribution is a mathematical and methodological synthesis with an explicit route from an observable's composition rule to a prediction that can fail. The classical functional equations, their stability theory, regular variation and the earlier transport models supply the starting points. The paper brings them together in three steps:

1. **Declare the equation for the observable.** The input and output operations, variable pair, normalisation and regularity conditions are stated before a family is assigned. Exact, approximate and asymptotic composition are kept distinct. A saturation model supplies additional dynamics; boundedness alone does not put it in the Cauchy grid (Sections 3 to 5).
2. **Expose the assumptions selecting the exponent.** The flow-accounting identity and conditional model of Section 6 state how demand and route length scale. The expression $d/(d + 1)$ is recovered under particular choices of those exponents. Velocity and the bridge from transport cost to biological mass are separate premises. Section 10 states the geometric bound only for the observable pair and assumptions that support it.
3. **Test the non-definitional premises and the resulting prediction.** The prospective transport test in Section 14 separates calibration of the proposed mechanism from held-out response measurements. The source comparisons ask whether the declared premises apply. The cosmological example distinguishes a change of variables from a physical explanation (Section 9); the recursion model keeps an assumed gain equation separate from a measured capability trajectory (Section 11).

This route gives the programme a way to formulate and audit particular scaling claims. It does not establish the empirical propositions of Law I, nor does the occurrence of the same exponent in another domain validate them. Section 16 gives the registered programme's status. Appendix E places each current contribution beside the closest earlier work used here; Appendix B preserves the dated record, and Appendix A records the earlier wording and its corrections.

2.3 The strongest objection: assumptions chosen after the fit

The strongest objection is that the framework could recover a familiar exponent by choosing its operators, dimension or transport assumptions after seeing the data. If a fitted slope is inverted to assign a dimension, or an inventory is calculated from the identity that the study is meant to test, the apparent agreement adds no empirical information. A correct conditional theorem can still be inapplicable to the system being measured.

The answer must be a measurement procedure, not a broader sufficiency claim. The system, variables and eligible network class must be declared in advance. The demand, route-length and velocity assumptions must be checked from measurements that do not merely reproduce the response being predicted, and the prediction must be assessed on held-out networks with independently measured inventory. A precise discrepancy under the declared premises counts against that model's applicability; failure of a premise is reported as a limit on its domain. The protocol must state these decisions before outcomes are examined. Until such a test is completed, the new empirical contribution remains prospective.

The narrower mathematical result is already available: once the flow model's explicit premises hold, its exponent follows. It replaces the earlier assertion that multiplicative composition, space filling and conservation, without further specification, must force $d/(d+1)$. The comparison with prior models is then a concrete question about their assumptions and observables, rather than a claim that different physics has been proved to obey one universal mechanism.

3 Name the function and its variables

A scaling statement identifies two quantities and a range. The same system can have a power law in one variable pair and an exponential in another. The following examples describe functions; they assign no physical mechanism by themselves.

Definition 1 (Power law)**.** For $x > 0$ and $A > 0$,

$$y = Ax^b, \qquad b \in \mathbb{R}. \tag{1}$$

Positive exponents below one are sublinear; one is linear; exponents above one are superlinear. Negative powers decrease. These are all power laws.

Definition 2 (Exponential)**.**

$$y = Ae^{\lambda x}. \tag{2}$$

Equal additive increments in x multiply the output by the same factor. This is distinct from a superlinear power law.

Definition 3 (One saturating response)**.** For $x \geq 0$, $K > 0$ and $Y_{\max} > 0$,

$$y = Y_{\max}\frac{x}{K + x}. \tag{3}$$

This is one bounded increasing form, not the unique consequence of a physical limit. Other bounded dynamics can converge differently or oscillate.

The logarithmic form belongs to the fourth operation pair in Section 4. Saturation requires a separate dynamical or response model. Choosing among these forms requires a declared observable, normalisation, error model and test; a visual resemblance is insufficient.

4 What the composition equations establish

A scaling explanation must specify what is combined and what the observable preserves. Classical functional equations give an exact answer once those choices are fixed (Cauchy 1821; Aczél 1966). Write $G_+ = (\mathbb{R}, +)$ for the additive real numbers and $G_\times = (\mathbb{R}_{>0}, \times)$ for the multiplicative positive real numbers. A homomorphism is a map that preserves the declared source and target operations.

Theorem 1 (Four continuous homomorphism problems)**.** *For each of the following source and target pairs, every continuous homomorphism has the stated form for some $c \in \mathbb{R}$, and each stated function is a homomorphism:*

1. *$F : G_+ \to G_+$ preserves addition in both variables:*

$$F(x + y) = F(x) + F(y), \qquad F(x) = cx, \qquad x, y \in \mathbb{R}. \tag{4}$$

2. *$F : G_\times \to G_\times$ preserves multiplication in both variables:*

$$F(xy) = F(x)F(y), \qquad F(x) = x^c, \qquad x, y > 0. \tag{5}$$

3. *$F : G_+ \to G_\times$ takes input addition to output multiplication:*

$$F(x + y) = F(x)F(y), \qquad F(x) = e^{cx}, \qquad x, y \in \mathbb{R}. \tag{6}$$

4. *$F : G_\times \to G_+$ takes input multiplication to output addition:*

$$F(xy) = F(x) + F(y), \qquad F(x) = c \log x, \qquad x, y > 0. \tag{7}$$

The logarithm is natural. The additive target is real-valued, not restricted to positive outputs. The constant cases are included: $c = 0$ gives the constant zero map into G_+ and the constant one map into $G_\times$. A zero-valued function is not a map into $G_\times$.

4.1 The proof and the scope of the four cells

Proof. First suppose $H : \mathbb{R} \to \mathbb{R}$ is continuous and additive. The identity at zero gives $H(0) = 0$; applying it to $x + (-x)$ gives $H(-x) = -H(x)$. Repeated addition gives $H(nx) = nH(x)$ for every integer n. For a positive integer n, $nH(1/n) = H(1)$, so $H(m/n) = (m/n)H(1)$ for every integer m. Thus $H(q) = cq$ on the rationals, where $c = H(1)$. For any real x, choose rationals $q_j \to x$. Continuity gives $H(x) = \lim_j H(q_j) = cx$.

This proves the first row. For the multiplicative row define $H(t) = \log F(e^t)$ for all real t. Positivity makes the logarithm defined, and the identity for F gives $H(s + t) = H(s) + H(t)$. Continuity therefore gives $H(t) = ct$, and substituting $t = \log x$ gives $F(x) = x^c$.

For the third row, $H(t) = \log F(t)$ is continuous and additive on $\mathbb{R}$, hence $F(t) = e^{ct}$. For the fourth, $H(t) = F(e^t)$ is continuous and additive, hence $F(x) = c \log x$. Direct substitution verifies every converse, including the constant cases. □

Four counts the chosen source and target operation pairs. It does not count all possible physical laws or four disjoint sets of numerical functions. The power-law exponent may be negative, zero, sub-linear or super-linear. A critical power law can have the power-law form; this theorem neither excludes it nor derives its critical exponent or physical mechanism. Other operations and other domains pose different classification problems.

Normalisation is part of the claim. If $Y(x) = Ax^c$ with $A > 0$, then $Y(xy) = Y(x)Y(y)/A$. Its normalised response $F(x) = Y(x)/Y(1)$ satisfies Equation (5). For an exponential measured from a first-pass baseline, $G(R) = e^{c(R-1)}$ has $G(1) = 1$; its additive coordinate is $r = R - 1$. For a baseline-one logarithmic response $G(R) = 1 + c \log R$, the additive output is $G - 1$, not G itself. Changing a coordinate or baseline can change the equation being tested.

4.2 A precise consequence for approximate composition

The following statement applies additive Hyers stability in declared logarithmic coordinates (Hyers 1941). It specifies both the error norm and its domain.

Proposition 1 (Global logarithmic-defect bound)**.** *Let $F : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ be continuous. If a constant $\varepsilon \geq 0$ satisfies*

$$|\log F(xy) - \log F(x) - \log F(y)| \leq \varepsilon \quad \textit{for every } x, y > 0, \tag{8}$$

then there is a unique $c \in \mathbb{R}$ for which

$$|\log F(x) - c \log x| \leq \varepsilon, \qquad e^{-\varepsilon} x^c \leq F(x) \leq e^{\varepsilon} x^c \quad (x > 0). \tag{9}$$

Proof. Put $h(t) = \log F(e^t)$ and $h_n(t) = 2^{-n} h(2^n t)$. The hypothesis gives

$$|h_{n+1}(t) - h_n(t)| \leq 2^{-n-1}\varepsilon$$

for every real t. Summing this bound shows that h_n converges uniformly to a continuous function A, with $|A(t) - h(t)| \leq \varepsilon$. Also

$$|h_n(s + t) - h_n(s) - h_n(t)| \leq 2^{-n}\varepsilon.$$

Passing to the limit proves that A is additive. The rational and continuity argument above gives $A(t) = ct$. If two constants c_1, c_2 met the uniform bound, then $|(c_1 - c_2)t| \leq 2\varepsilon$ for every real t, forcing $c_1 = c_2$. Substitution of $t = \log x$ and exponentiation give the result. □

An absolute residual $|F(xy) - F(x)F(y)|$ is a different error metric. Observations of finitely many noisy pairs cannot establish the global bound in Equation (8). The proposition contains no dynamics that attract a system towards a family. Its conclusion is proximity under a stated hypothesis, not a claim that perturbations in nature must take this form.

4.3 Scaling limits and regular variation

Suppose F is positive and measurable on the positive reals and, for every $x > 0$, the finite positive limit

$$H(x) = \lim_{t \to \infty} \frac{F(tx)}{F(t)} \tag{10}$$

exists. Then

$$\frac{F(txy)}{F(t)} = \frac{F(txy)}{F(tx)} \frac{F(tx)}{F(t)}$$

Table 1. Assumptions that must accompany a composition claim.

Choice	What must be stated
Observable and coordinates	Input and output variables, their domains, units and baseline
Operations	Both the input and output operation; an update rule alone supplies neither identity
Regularity	The continuity assumption used above, or the precise replacement theorem
Approximation	The norm, quantified domain and error bound, rather than a finite fit alone
Scaling limit	Which ratios converge, over which range, and with what residual variation
Physical interpretation	The separate mechanism and measurement linking the mathematical objects to a system

gives $H(xy) = H(x)H(y)$, because $tx \to \infty$ for fixed $x > 0$. The limit is measurable, for example by taking the sequence $t = n$. The classical measurable Cauchy result then gives $H(x) = x^c$. Writing $\mathcal{L}(t) = F(t)/t^c$ yields

$$F(t) = t^c \mathcal{L}(t), \qquad \frac{\mathcal{L}(tx)}{\mathcal{L}(t)} \longrightarrow 1 \quad \text{for every fixed } x > 0.$$

This is the regular-variation characterisation, with $\mathcal{L}$ a slowly varying factor (Bingham et al. 1987). An asymptotic power ratio need not be an exact power at finite scale. For example, at large t, $t^c \log t$ has the same limiting power ratios but local logarithmic slope $c + 1/\log t$.

The existence and positivity of the limits are hypotheses, not conclusions about every measured curve. Generalised functional equations and homomorphisms form part of the established setting for these questions (Ostaszewski 2016). The empirical contribution sought here is to justify the equation or limit before inspecting the validation curve, then test the resulting prediction against alternatives.

5 The principle and its empirical test

A recursive update describes a succession of states, $Q_{n+1} = T(Q_n, u_n)$, where u_n names any additional input. A composition equation instead relates measurements made at combined inputs. To obtain a scaling prediction, a model must connect the update procedure to the observable equation. Dependence on an earlier state, or associativity of a chosen operation, alone does not provide that connection.

The two operations must both be named. Multiplication of input scale ratios together with multiplication of normalised outputs gives the power-law row of Theorem 1. Addition of input intervals together with multiplication of positive outputs gives its exponential row. Addition of both inputs and outputs gives a linear map; multiplication of inputs together with addition of real outputs gives a logarithm. These are statements about observables, not labels that a substrate acquires merely by branching or feeding back.

For example, the state rule $Q_{n+1} = qQ_n$ with $q > 0$ and $Q_0 > 0$ at integer ticks gives $Q_n = Q_0 q^n$. Although the state is multiplied at every update, the input tick count adds, and the normalised output is exponential in that count. Calling the update "multiplicative" without identifying the input operation would give the wrong classification.

A bound and convergence are different premises. A bounded sequence can oscillate: $Q_n = 2 + (-1)^n$ never converges. For the particular capped update

$$Q_{n+1} = \min(Q_n + \delta_n, K), \qquad 0 \le Q_0 \le K, \quad \delta_n \ge 0,$$

monotonicity and boundedness together imply convergence to a limit at most K. In fact $Q_n = \min(Q_0 + \sum_{j<n} \delta_j, K)$. Its limit equals K only when the cumulative increments reach or exceed the remaining gap; finite-time attainment requires a finite partial sum to do so. Different bounded dynamics can approach a limit at different rates, or fail to approach any limit. Saturation therefore requires its own dynamics and is not a fifth Cauchy cell.

The ARC Principle. The ARC Principle remains Law I of the ARC Theory, a law under test, with its statement and verdict rules fixed by the theory registration. This paper asks whether a justified composition description can predict later scaling. The classical theorem supplies conditional forms; a scientific test must establish why the relevant observable obeys the proposed equation and whether that description predicts held-out observations better than rivals. Neither the theorem nor a retrospective fit supplies a registered verdict. The measurement requirements are set out in Section 15.

6 Where the network exponent comes from

A network's dimension describes its geometry. To predict how much it transports, we also need the demand it serves and the distance over which it carries that demand. The following model separates these quantities. It gives a conditional derivation of the familiar fraction $d/(d+1)$ and identifies the additional measurement needed before that fraction can describe metabolism against body mass.

6.1 An exact accounting identity

Consider a finite rooted tree that carries a resource from one source to sinks i. Each sink receives a steady non-negative rate q_i, with at least one $q_i > 0$. Edge e has length $\ell_e \geq 0$ and carries the sum of the demands downstream of it:

$$Q_e = \sum_{i : e \in P_i} q_i, \tag{11}$$

where P_i is the route from the source to sink i. Flow is conserved at intermediate junctions; the resource counted here is neither created nor consumed before reaching its designated sink. Define total throughput B, flow-length cost C and the demand-weighted mean route length $\bar{\ell}_q$ by

$$B = \sum_i q_i, \qquad C = \sum_e \ell_e Q_e, \qquad \bar{\ell}_q = \frac{\sum_i q_i \ell(P_i)}{B}. \tag{12}$$

If q_i is measured in resource amount per unit time, C has units of resource amount times distance per unit time. It is a flow-length cost, not automatically energy expenditure, network mass or body mass.

Lemma 1 (Flow accounting)**.** *Under these definitions,*

$$C = B\bar{\ell}_q. \tag{13}$$

Proof. Each sink's demand contributes to every edge on its route. Exchanging the two finite sums gives

$$\begin{aligned} C &= \sum_e \ell_e \sum_{i : e \in P_i} q_i = \sum_i q_i \sum_{e \in P_i} \ell_e \\ &= \sum_i q_i \ell(P_i) = B\bar{\ell}_q. \end{aligned} \tag{14}$$

□

The same accounting applies to a network with split routes when every non-negative route flow and every traversal is specified. If there is a circulating flow in addition to the source-to-sink routes, its contribution must be retained:

$$C = B\bar{\ell}_q + C_{\mathrm{circ}}, \qquad C_{\mathrm{circ}} \geq 0. \tag{15}$$

Here $\bar{\ell}_q$ refers only to the specified source-to-sink routes. Alternatively, recirculation that belongs to a delivered resource's itinerary can be included in that itinerary's full length. Counting only its shortest path while charging the actual circulating edge flows would mix two different models.

6.2 Demand and route length select the exponent

Let L denote a dimensionless size coordinate, such as the ratio of a domain's linear extent to a fixed reference length. All dimensional scales below are carried by positive constants B_0 and ℓ_0.

Proposition 2 (Conditional throughput scaling)**.** *For a family satisfying Eq. (13), suppose that, as $L \to \infty$,*

$$\frac{B(L)}{B_0} \sim L^{\mu}, \qquad \frac{\bar{\ell}_q(L)}{\ell_0} \sim L^{\zeta}, \qquad \mu > 0, \quad \mu + \zeta > 0. \tag{16}$$

With $C_0 = B_0\ell_0$, throughput against flow-length cost obeys

$$\frac{B}{B_0} \sim \left(\frac{C}{C_0}\right)^{\alpha_C}, \qquad \alpha_C = \frac{\mu}{\mu + \zeta}. \tag{17}$$

Proof. The accounting identity gives $C/C_0 \sim L^{\mu+\zeta}$. Raising this relation to $\mu/(\mu + \zeta)$ and dividing by $B/B_0 \sim L^{\mu}$ gives a ratio tending to one. □

An asymptotic exponent is not a finite-range prediction. For dimensionless $L > 0$ and a constant $A > 0$, take $B/B_0 = L + A$ and $\bar{\ell}_q/\ell_0 = L$. Then $C/C_0 = L(L + A)$, and both input scalings have limiting exponent one. The asymptotic throughput exponent against C is $1/2$, but direct differentiation gives the local slope

$$\frac{\mathrm{d}\log B}{\mathrm{d}\log C} = \frac{L}{2L + A} < \frac{1}{2}. \tag{18}$$

A finite-range fit need not recover the limit. A predictive experiment must therefore lock either the complete finite-range functions and their uncertainty or explicit bounds on the finite-size corrections. Simply inserting two limiting exponents into Equation (17) does not supply that prediction.

The geometric special case is therefore

$$\alpha_C = \frac{d}{d + 1} \qquad \text{when } \mu = d \text{ and } \zeta = 1. \tag{19}$$

Both premises need a reason. Suppose a d-dimensional family has a number of sinks proportional to L^d and a mean demand per sink tending to a positive size-independent value. Then total demand has exponent $\mu = d$. Constant sink density in an expanding domain is one way to obtain the required sink count. A change in density or demand per sink changes this conclusion even when the embedding dimension stays fixed.

Route scaling needs an equally explicit condition. Suppose a fixed positive fraction of total demand lies at distance at least $cL\ell_*$ from the source, where $c > 0$ and ℓ_* is a fixed length, while every served route has length at most $KL\ell_*$ for a size-independent K. The lower distance condition and the upper route bound place $\bar{\ell}_q$ between positive constant multiples of L. Within the power-scaling assumptions of Eq. (16), they force $\zeta = 1$. Bounds alone do not supply a limiting prefactor; the stronger asymptotic relation also requires

convergence of the normalised mean route length, for example through a specified self-similar family. Demand concentrated near the source, increasing path stretch or unaccounted recirculation changes the premises.

The demand-distance argument also gives a conditional bound. If the positive fraction at distance at least $cL\ell_*$ is retained but the upper route bound is dropped, then $\bar{\ell}_q$ is at least a positive constant times L. Under Eq. (16), $\zeta \geq 1$. With $\mu = d > 0$, this implies $\alpha_C \leq d/(d+1)$. The bound concerns the declared throughput and flow-length cost; its hypotheses are part of the result.

6.3 Inventory and body mass are separate quantities

An inventory relation can be derived when the transported resource moves at a common speed v on every edge. In steady flow, the amount in transit on edge e is its rate Q_e multiplied by its transit time ℓ_e/v. Thus

$$N = \sum_e Q_e \frac{\ell_e}{v} = \frac{C}{v}. \tag{20}$$

Here N counts the resource in transit. It excludes stored material and waiting time unless those are added to the residence-time accounting. If edge speeds differ, the exact expression is instead $N = \sum_e Q_e \ell_e / v_e$; an arbitrary average speed cannot replace the v_e.

A speed that is uniform within each network may still change between sizes. If $v/v_0 \sim L^\upsilon$, then

$$\frac{N}{N_0} \sim L^{\mu+\zeta-\upsilon}, \qquad \alpha_N = \frac{\mu}{\mu+\zeta-\upsilon}, \qquad N_0 = \frac{C_0}{v_0}, \tag{21}$$

where α_N is the throughput exponent against inventory and the denominator must be positive. Constant speed across sizes is the special case $\upsilon = 0$.

Body mass M requires its own bridge. If independent physiological evidence establishes

$$\frac{M}{M_0} \sim \left(\frac{C}{C_0}\right)^\eta, \qquad \eta > 0, \tag{22}$$

then the metabolic interpretation of B, if established, would give the mass exponent

$$\alpha_M = \frac{\mu}{\eta(\mu+\zeta)}. \tag{23}$$

The constants in Eq. (22) carry the units. Neither $\eta = 1$ nor a proportionality between body mass and inventory follows from geometric dimension. A measured metabolic quantity must also correspond to the resource throughput being modelled.

The velocity dependence is explicit in Banavar et al. (2010). Their radial and hierarchically branched models supply a three-dimensional volume. They obtain a metabolic mass exponent of $2/3$ when flow velocity is independent of body size, and an upper value of $3/4$ when velocity scales as $M^{1/12}$ under their additional service-volume constraints. Their assumptions make the inventory in transit proportional to mass and route length proportional to $M^{1/3}$. In that setting the accounting reads

$$B \propto \frac{Mv}{\bar{\ell}_q} \propto M^{2/3+\nu}, \qquad v \propto M^\nu. \tag{24}$$

This explains the two exponents without changing the spatial dimension. The permissible upper value of ν is supplied by that model's constraints, not by the accounting identity. Its demand and mass assumptions must not be silently replaced by $\mu = d$ and $M \propto C$.

6.4 What a cross-domain comparison tests

The useful prediction is now explicit: independently characterised demand and routing determine a throughput-cost exponent, and a separately justified bridge determines whether it is also a mass exponent. A comparison must state the two measured variables, the size range, physiological or physical conditions, and the uncertainty in each premise. Computing C from $B\bar{\ell}_q$ and then recovering the same identity is an accounting check; evidence for the model comes from independent measurements of the premises and a prediction of outcomes not used to choose them.

Biological measurements distinguish these possibilities. White and Seymour (2003) report mammalian basal-metabolism fits compatible with $2/3$ after temperature adjustment and restrictions on eligible measurements. Kolokotrones et al. (2010) report curvature in mammalian metabolic scaling, so a fitted constant exponent depends on the mass range. These results concern defined datasets and measurement conditions. Source-specific values and methods belong with those definitions in Section 12; selected means do not supply a new validation of Eq. (19).

The same requirement applies outside biology. A roughness exponent, a distribution-tail exponent, a throughput exponent and a cosmological scale-factor exponent describe different variable pairs. An analytic equality between selected fractions supplies no measurement of demand or routing. Section 9 gives a complete example: its powers follow from the fluid equations, while the apparent network dimension is introduced afterwards by a change of variables. The transport result remains a definite conditional statement, with a mechanism and observables that can be specified before a test.

7 What metabolic scaling implies for the heartbeat

The heart supplies oxygen by moving blood, but a beat is not a fixed quantity of oxygen delivery across animals. Stroke volume and the oxygen extracted from each unit of blood connect metabolic demand to heart rate. Writing that balance explicitly shows what is required for a quarter-power relation.

7.1 The oxygen balance

Let $\dot{V}_{\mathrm{O}_2}$ denote whole-body oxygen consumption per unit time, f_H heart rate, V_s stroke volume and ΔO_2 the difference between arterial and mixed-venous oxygen content per unit blood volume. At steady state, with no net accumulation of the oxygen being accounted for, consumption equals blood flow multiplied by the oxygen removed from that flow. Since cardiac output is $f_H V_s$, the Fick balance is

$$\dot{V}_{\mathrm{O}_2} = f_H V_s \Delta O_2. \tag{25}$$

The units are oxygen amount per unit time on each side: beats per unit time, multiplied by blood volume per beat and oxygen amount per blood volume. This is a conservation relation, not a spatial scaling law.

Use the dimensionless mass $m = M/M_0$ and suppose, over a declared range and comparable physiological conditions, that

$$\dot{V}_{\mathrm{O}_2} = \dot{V}_0 m^b, \qquad V_s = V_{s0} m^s, \qquad \Delta O_2 = \Delta O_{20} m^e, \tag{26}$$

with positive prefactors. Substitution gives

$$f_H = f_{H0} m^h, \qquad h = b - s - e, \qquad f_{H0} = \frac{\dot{V}_0}{V_{s0}\Delta O_{20}}. \tag{27}$$

An asymptotic version follows if the three input relations are asymptotic rather than exact powers. In either case, the familiar $h = -1/4$ follows from the three scaling assumptions $b = 3/4$, $s = 1$ and $e = 0$.

The oxygen consumed grows as the three-quarter power, stroke volume grows in proportion to mass, and oxygen extraction per unit blood volume is independent of mass. A metabolic-energy exponent can stand in for b only when its conversion to oxygen consumption is justified for the measurements in question.

Mass-specific oxygen consumption is a different quantity:

$$\frac{\dot{V}_{O_2}}{M} = \frac{\dot{V}_0}{M_0} m^{b-1}. \tag{28}$$

Its exponent equals the heart-rate exponent only if $s + e = 1$. The balance therefore gives a testable relation between three physiological scalings; dimension alone supplies neither an individual's pulse nor its prefactor.

7.2 Physiological state is part of the observation

Goldbogen et al. (2019) recorded one male blue whale for 8.5 hours. Diving heart rates were typically 4 to 8 beats per minute, with a minimum of 2; post-dive surface rates reached 25 to 37. Their approximately 15 beats per minute resting value was an allometric prediction, not a direct measurement of a resting constant. These observations show why diving, recovery and resting values must retain their conditions. They are not interchangeable points in a cross-species resting-heart-rate fit.

A comparison between species must therefore state the activity and thermal conditions, mass measurements, reference ranges and uncertainty, and must establish the stroke-volume and extraction assumptions relevant to that comparison. Choosing one representative pulse for each animal and supplying one animal's value as a prefactor does not independently test a prediction from network dimension.

7.3 Lifetime totals require a lifetime average

The number of beats over a lifespan τ is the integral of heart rate over that life:

$$N_{\text{beats}} = \int_0^\tau f_H(t)\,\mathrm{d}t = \bar{f}_H \tau, \qquad \bar{f}_H = \frac{1}{\tau}\int_0^\tau f_H(t)\,\mathrm{d}t. \tag{29}$$

Here $\bar{f}_H$ is the lifetime mean, not a resting measurement. If that mean and a consistently defined lifespan scale as $m^{\bar{h}}$ and m^{ℓ}, their product scales as $m^{\bar{h}+\ell}$. Cancellation requires $\bar{h} + \ell = 0$. It removes a mass exponent within the specified model; it does not establish a universal numerical total across species, activity histories or definitions of lifespan. The prefactors and the replacement of a lifetime mean by a convenient resting value remain separate empirical questions.

8 The fission chain reaction

The rest-energy relation $E = mc^2$ (Einstein 1905) and the kinetics of a chain reaction answer different questions. A reaction's energy release depends on the difference between initial and final states. In a simplified branching description, the expected number of events in generation n obeys $N_{n+1} = kN_n$, with a fixed effective multiplication factor k. Then $N_n = N_0 k^n$ is exponential in additive generation count. This is a mathematical description under a fixed-factor premise, not a claim that an actual transient keeps that factor constant.

What the grid does and does not explain: the chain reaction. A branching model grows exponentially while its effective multiplication factor exceeds one and stays constant. That is a branching process, and the arithmetic was understood in the late 1930s; it owes nothing to this work. In the four-cell grid it

Table 2. The chain reaction's regimes.

Regime	k	Cell of the grid	Behaviour	Status
Subcritical	$k < 1$	Additive input, multiplicative output	Exponential decay	Classified, not predicted
Supercritical	$k > 1$	Additive input, multiplicative output	Exponential growth	Classified, not predicted
Controlled	$k \approx 1$	Bounded by feedback (outside the grid)	Steady power	Classified, not predicted

sits in the additive-input, multiplicative-output cell, so the grid *classifies* it correctly. Classifying is not explaining. The physical mechanism and the conditions that sustain its effective factor must be supplied separately; naming the cell adds no physical content.

The box this section carried until 11 August 2026 is recorded in Appendix A.3.

9 Cosmic expansion and the meaning of a shared formula

Cosmology provides a precise example of the difference between deriving a power and relabelling it. The expansion law follows from gravitational dynamics and the fluid's equation of state. Expressing its exponent as $d/(d+1)$ afterwards does not supply an independently measured transport dimension.

9.1 The fluid equations determine the expansion

Consider an expanding, spatially flat, homogeneous and isotropic model with one positive-density perfect fluid and constant equation-of-state parameter w. Write $P = w\rho c^2$, with ρ the mass-equivalent density, P pressure and c the speed of light. There is no additional component in this single-fluid solution. With $H = \dot{a}/a$, the Friedmann and continuity equations give (Carroll et al. 2003)

$$H^2 = \frac{8\pi G}{3}\rho, \tag{30}$$

$$\dot{\rho} = -3H(1+w)\rho, \tag{31}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(1+3w)\rho. \tag{32}$$

For $w > -1$, integrating on the expanding branch gives

$$a(t) = a_* \left(\frac{t - t_0}{t_*}\right)^p, \qquad p = \frac{2}{3(1+w)}, \qquad t > t_0, \tag{33}$$

where $a_* > 0$ and $t_* > 0$ are reference scales consistent with the density normalisation. One way to see the exponent is to integrate Eq. (31), obtaining $\rho \propto a^{-3(1+w)}$, and insert this into Eq. (30). The resulting equation makes $a^{3(1+w)/2}$ proportional to $t - t_0$.

The case $w = -1$ is solved separately. Its density is constant, so the expanding branch has constant $H = H_\Lambda > 0$ and

$$a(t) = a_* \exp\big[H_\Lambda (t - t_*)\big]. \tag{34}$$

This exponential follows by integrating $\dot{a} = H_\Lambda a$. It is not a finite power of time obtained by substituting $w = -1$ into Eq. (33).

9.2 Acceleration and exponential expansion are distinct

The deceleration parameter of the single-fluid model is

$$q_{\mathrm{dec}} = -\frac{a\ddot{a}}{\dot{a}^2} = \frac{1+3w}{2}. \tag{35}$$

Positive density therefore gives acceleration when $w < -1/3$. At $w = -1/3$, the power exponent is $p = 1$ and the solution coasts linearly. For $-1 < w < -1/3$, it is an accelerating power law. In particular, $w = -2/3$ gives $p = 2$ and $a(t) \propto (t - t_0)^2$. Exponential expansion occurs at $w = -1$ under the assumptions above (Carroll et al. 2003).

Version 3.13 already corrected the earlier identification of the acceleration boundary with a change from power-law to exponential expansion. That correction is retained here. The quadratic example makes its consequence explicit: acceleration does not require an exponential. A linear function is itself a member of the power-law family, not a boundary of that family's mathematical existence.

9.3 An algebraic coordinate, not a measured dimension

For a finite power exponent $p \neq 1$, define

$$d_{\mathrm{eff}} = \frac{p}{1-p} = \frac{2}{1+3w}, \qquad p = \frac{d_{\mathrm{eff}}}{d_{\mathrm{eff}}+1}. \tag{36}$$

In this section d_{eff} is solely the coordinate defined by that equation. It is not the independently specified geometric d in Section 6. Within the expanding power-law branch, the map is finite for $w > -1$ except at $w = -1/3$. It is positive for $w > -1/3$ and less than -1 for $-1 < w < -1/3$.

At $w = -1/3$, the map has a pole while the physical solution remains linear and regular. Approaching the boundary from the decelerating side sends d_{eff} to positive infinity; approaching from the accelerating side sends it to negative infinity. Formally substituting $w = -1$ gives $d_{\mathrm{eff}} = -1$, where the expression for p is undefined. The exponential solution therefore has no finite power exponent represented by this coordinate.

Table 3. Analytic cases of the flat, constant, single-fluid model. The entries follow from its equations; they are not measured network dimensions or independent confirmations of a network law.

Fluid or case	w	Time exponent p	q_{dec}	d_{eff}
Stiff fluid	1	1/3	2	1/2
Radiation	1/3	1/2	1	1
Pressureless matter	0	2/3	1/2	2
Coasting	−1/3	1	0	Undefined
Accelerating example	−2/3	2	−1/2	−2
Vacuum	−1	Exponential	−1	Not applicable

The radiation and pressureless-matter powers follow from $w = 1/3$ and $w = 0$ in the fluid equations. Those equations assume a homogeneous background; neither a photon-path network nor a cosmic-web partition enters their derivation. Interpreting the numbers 1 and 2 in the last column as such mechanisms would add a premise that has not been established by the change of variables.

9.4 Idealised components and cosmic history

A single constant w is an idealisation, not a complete history of a universe containing several evolving contributions. The powers in Table 3 describe the stated one-component models and can describe limiting

regimes when their assumptions apply. A mixture, an evolving equation of state, spatial curvature or the branch with $w < -1$ requires its own solution. A change in the dominant component does not, by itself, establish a change in an independently measured composition operator.

9.5 What the common form establishes

Equation (36) is a reparametrisation of an exponent already derived from cosmology. It supplies no extra equation for the expansion, and fitting d_{eff} from the same power cannot be an independent prediction of that power. In the transport calculation, by contrast, the proposed exponent depends on demand, routes and an observable bridge that must be specified before their prediction is tested.

The connection that survives is exact and useful: a shared functional form can organise a comparison while leaving the physical derivations distinct. A claim of a common mechanism would require additional equations and independently identified quantities linking the two domains. The fluid calculation itself supplies no biological prediction, and a biological exponent is not a measurement of the cosmological coordinate in Eq. (36).

10 A conditional geometric bound

Geometry becomes predictive when the transported quantity and the route assumptions are specified. Use the source-to-sink model of Section 6: sink i receives a nonnegative demand rate q_i, with positive total demand $B = \sum_i q_i$; edge e has length ℓ_e and downstream flow Q_e. For the declared routes P_i, the flow-weighted mean path length is $\bar{\ell}_q = B^{-1} \sum_i q_i \ell(P_i)$. On a rooted tree, exchanging the sums gives

$$C = \sum_e \ell_e Q_e = \sum_i q_i \ell(P_i) = B\bar{\ell}_q. \tag{37}$$

Here C has units of demand rate times length. It is not automatically energy expenditure, material inventory or body mass. A network with recirculation needs its additional transported flow accounted for explicitly.

Consider a family indexed by dimensionless linear scale $L \to \infty$, and suppose

$$B(L) \sim b_0 L^{\mu}, \qquad \bar{\ell}_q(L) \sim \ell_0 L^{\zeta}, \qquad b_0, \ell_0 > 0, \quad \mu > 0. \tag{38}$$

When $\mu + \zeta > 0$, Equation (37) implies

$$B \sim b_0^{1-\alpha_C} \ell_0^{-\alpha_C} C^{\alpha_C}, \qquad \alpha_C = \frac{\mu}{\mu + \zeta}. \tag{39}$$

The exponent concerns this observable pair. It is not selected by the Cauchy equation alone.

Proposition 3 (Demand at macroscopic distance)**.** *Suppose fixed constants $0 < p \le 1$ and $c > 0$, independent of L, have the following property for all sufficiently large networks: sinks at Euclidean distance at least cL from the source account for at least the fraction p of total demand. Under Equation (38), route lengths measured in the same physical metric obey $\zeta \ge 1$. Consequently*

$$0 < \alpha_C \le \frac{\mu}{\mu + 1} < 1.$$

Proof. Every route to those sinks has length at least their source-to-sink distance. Hence $\bar{\ell}_q \ge pcL$. If $\zeta < 1$, the assumed relation $\bar{\ell}_q \sim \ell_0 L^{\zeta}$ would give $\bar{\ell}_q / L \to 0$, contradicting this lower bound. Thus $\zeta \ge 1$, and the inequality follows from $\mu > 0$. □

Setting $\mu = d$ needs an additional demand assumption. For example, if the number of sinks is asymptotic to $n_0 L^d$ under fixed sink density and their average demand tends to a positive constant q_0, then $B \sim n_0 q_0 L^d$. Embedding a network in a d-dimensional domain alone does not fix its sink number or demand. If, in addition to the lower bound above, route stretch is bounded so that $\bar{\ell}_q \leq KL$ with fixed K, the stated power asymptotic has $\zeta = 1$. Only with both choices does $\alpha_C = d/(d+1)$ follow as an equality.

Table 4. Conditional throughput versus transport exponents when $\mu = d$ and $\zeta = 1$; these are analytic values, not measurements.

d	$\alpha_C = d/(d+1)$	$1 - \alpha_C$
1	1/2	1/2
2	2/3	1/3
3	3/4	1/4
10	10/11	1/11

To transfer this result to another response variable requires another model. If body mass satisfies $M \sim m_0 C^\eta$ with $m_0, \eta > 0$, the corresponding exponent is $b_M = \alpha_C/\eta$, not automatically α_C. At a specified common transport velocity v, inventory in transit is C/v; scale-dependent velocity changes that bridge. The physical assumptions in transport explanations must therefore be compared directly (West et al. 1997; Banavar et al. 1999, 2010). An observed exponent above one for a different variable pair contradicts neither Proposition 3 nor finite-dimensional geometry. The limit $d/(d+1) \to 1$ as $d \to \infty$ is a limit of this formula, not a physical phase boundary.

11 Recursive capability and correction

The transport exponent of Section 10 concerns demand against a flow-weighted transport quantity. Capability against recursive depth is another observable pair. A capability exponent above one would not be an escape from a physical law; it would require its own model and measurement, with the hardware, energy, latency and evaluation costs still accounted for.

11.1 A fixed rule can generate super-linear growth

The unchanged rule

$$x_{n+1} = (\sqrt{x_n} + 1)^2, \qquad x_0 = 1, \tag{40}$$

gives $x_n = (n+1)^2$ for every integer $n \geq 0$. The initial value has that form, and substituting $(n+1)^2$ into the rule gives $(n+2)^2$. This is a counterexample to the mathematical inference that fixed rules force sub-linearity; it is not an AI measurement.

Frozen model weights, fixed agent code, persistent memory, prompts, tools and the procedure that proposes subsequent improvements are separate experimental choices. A change to any one does not guarantee improvement or faster improvement in the next round. A test must identify what changes, what remains fixed, the available resources and the outcome. The error-decay fit reported in Paper II remains inconclusive, and its earlier super-linear estimate remains retracted; neither result is replaced by this counterexample.

11.2 The shifted cumulative-advantage model

Let $r \geq 0$ count added passes, with dimensionless amplification $g(0) = 1$. A separate dynamical assumption is

$$\frac{dg}{dr} = ag^{\beta_L}, \qquad a > 0, \quad \beta_L < 1. \tag{41}$$

At fixed parameters, integration gives

$$\frac{g^{1-\beta_L} - 1}{1 - \beta_L} = ar, \qquad g(r) = \left(1 + \frac{a}{\alpha} r\right)^{\alpha},$$

where

$$\alpha = \frac{1}{1 - \beta_L} > 0. \tag{42}$$

Thus $g(r) \sim (a/\alpha)^{\alpha} r^{\alpha}$ as $r \to \infty$. At $\beta_L = 0$ the marginal gain is constant and $g = 1 + ar$, an affine gain rather than an assertion of statistically independent steps. For $0 < \beta_L < 1$ the solution is convex and its asymptotic exponent exceeds one; negative β_L gives an asymptotic exponent below one. These are properties of the assumed ODE, not architectural classifications. A response such as task accuracy is bounded, so an unbounded gain law cannot describe it at arbitrary depth. Its measured regime, saturation and evaluation axis must be specified; a finite improvement in a bounded score is not evidence for an indefinitely valid asymptotic growth law.

For $r > 0$ the local elasticity is

$$\frac{d \log g}{d \log r} = \frac{\alpha r}{r + \alpha/a}. \tag{43}$$

The scale $r_c = \alpha/a$ is where that elasticity equals $\alpha/2$. It is not a discontinuity or, in general, the onset of elasticity above one. When $\alpha > 1$, that onset occurs at $r > r_c/(\alpha - 1)$. Over a finite window a fitted exponent need not equal α.

In first-pass coordinates put $R = r + 1$ and $G(R) = g(R - 1)$. For a fixed positive baseline I, the model reads

$$U(R) = IG(R), \qquad G(R) = \left[1 + \frac{a}{\alpha}(R - 1)\right]^{\alpha}, \qquad R \geq 1. \tag{44}$$

Here $G(1) = 1$. The exact identity $G(R) = R^{\alpha}$ requires $a = \alpha$; otherwise that power is an asymptotic form with its own prefactor. Elasticity against R is $\alpha R/(R - 1 + \alpha/a)$, a different finite-depth quantity from Equation (43).

There is an exact composition identity for the shifted solution, but its input operation is deformed addition:

$$r \oplus_a s = r + s + \frac{a}{\alpha} rs, \qquad g(r \oplus_a s) = g(r)g(s), \qquad r, s \geq 0. \tag{45}$$

Indeed $1 + (a/\alpha)(r \oplus_a s) = [1 + (a/\alpha)r][1 + (a/\alpha)s]$. This injective change of variable also proves associativity and gives the identity element zero on the nonnegative half-line. The resulting structure is a commutative monoid, not the full additive real group in Theorem 1. Generalised functional equations treat such distinctions (Ostaszewski 2016). Recognising the operation does not establish that a measured agent obeys it. Nor does the limit $\beta_L \to 1$ at fixed a, r, which gives $g(r) \to e^{ar}$, create a finite-depth divergence.

11.3 Elapsed time, correction and the conditional ceiling

Depth and time cannot be interchanged without a clock model. Paper XIII defines self-acceleration through cycle duration proportional to capability to the power $-\delta$, with constant $\delta \geq 0$ on a fixed substrate. With fixed parameters and that duration law, combining the gain ODE with elapsed time gives $k = \delta - 1/\alpha$

in a capability-growth law proportional to U^{1+k}. Hardware procurement, a proportional overhead and an irreducible additive serial latency are not the same clock change. An additive latency changes the local self-acceleration exponent; the constant-exponent clock law must therefore be checked rather than inferred from faster completion alone.

Paper X separately models departure magnitude D, the normalised index D/U, correction strength and capability growth. Its correction exponent, denoted β_C here, is not the gain coupling β_L. With positive gain drift, no level drift and the specified power-law correction model, $\beta_C > k$ is the condition for the modelled fraction to vanish. With positive gain and level drift the stated condition is $\beta_C > \max(k, 0)$. These results assume the model's forcing and correction laws; an unbounded power-law corrector is a substantive premise. At equality, and with compounding, coefficients and forcing matter. Delay, finite capacity and a change of measurement clock require their own analysis. A vanishing fraction can accompany growing absolute D, and neither quantity is itself a measure of harm. A regular equation in a depth coordinate does not guarantee enough wall-clock time for human intervention.

The registered conditional pacing relation of Law III, the ARC Ceiling, a law under test, is

$$\alpha_{\text{crit}} = \frac{1}{1-\gamma}, \qquad 0 \le \gamma < 1. \tag{46}$$

It requires the specified burden model and zero direct-depth term; it is not derived here from a joint model of capability, correction accumulation, covariance, delay and absolute harm. On the displayed domain, $\alpha_{\text{crit}} \le 2$ is equivalent to $\gamma \le 1/2$. The same-class inequality carried by P6 with P20, and the conjectured equality at one half, remain distinct from the ODE relation in Equation (42). Their similar algebra supplies no independent confirmation. The theory registration and its frozen verdict rules remain authoritative; this section adds no verdict or rule and establishes no universal ceiling of two.

11.4 Quantum error correction: distinguish the axes

The surface-code result reports logical error per cycle against code distance, with suppression factor $\Lambda = 2.14 \pm 0.02$ per distance increase of two (Google Quantum AI and Collaborators 2025). The preprint was public from 24 August 2024 and the result announced on 9 December 2024; its relation to this programme is convergent timing, not a prediction. Code distance, physical-qubit count and elapsed correction cycles are different inputs. The author correction published on 28 April 2026 changes the label of Figure 3a from surface-code distance to repetition-code distance and corrects its legend keys; it is not a change of that horizontal axis into elapsed cycles (Google Quantum AI and Collaborators 2025).

Table 5. Distinct quantum-error-correction comparisons and their assumptions.

Input axis	What the corresponding statement means
Code distance d_{code}	Below-threshold error suppression is reported against spatial redundancy, not recursion duration.
Physical-qubit count N	Under the illustrative relations $p_L = p_0 e^{-\kappa d_{\text{code}}}$ and $N = \nu d_{\text{code}}^2$, substitution gives $p_L = p_0 e^{-\kappa\sqrt{N/\nu}}$, not an exponential in N.
Cycle count T	For independent cycles with constant $0 < p_L < 1$, cumulative failure probability is $1 - (1 - p_L)^T$, tending to one as T increases. Temporal dependence needs a different model.

A stable per-cycle error rate therefore does not imply saturation of cumulative failure at a harmless ceiling. These are established or conditional error-accounting relationships, not proof of the proposed capability model. Likewise, a favourable capability or correction exponent demonstrates neither intrinsic

ethical behaviour nor a deployed safety guarantee. Evaluation must specify the values and tasks, measure usefulness alongside correction, and retain human oversight, containment and shutdown for as long as they can be held. Embedded correction is the programme's contingency for control lost against our will, whose effectiveness still requires the relevant tests.

12 Evidence, with its measurement and scope

The mathematical statements in Sections 4 and 6 follow from their premises. Applying them requires separate evidence for those premises. An analytic identity, a fitted slope and a prospectively tested prediction are different kinds of support.

12.1 Biological comparisons

The source-specific metabolic estimates and their measurement conditions are retained in Appendix F. They are not pooled into a new estimate of a universal exponent. In particular, the fungal table mixes biomass with radial extension; its former mean and tests do not establish one transport scaling relation. A non-significant test against one half is not an equivalence result. The earlier three-group ANOVA and model comparisons also relied on a group subsequently excluded from the proposed network class. Those comparisons are retained as superseded history and carry no current validation weight.

Note on the $\frac{3}{4}$ debate. The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method (White and Seymour 2003; Glazier 2005, 2022). White and Seymour's data set of 619 species gives an exponent of 0.69 and, once body temperature and digestive state are allowed for, exponents not significantly different from $\frac{2}{3}$ (White and Seymour 2003); Savage et al. (2004) find 0.737 for 626 species with the data binned by mass (0.712 unbinned); and Kolokotrones et al. (2010) find the relation convex on logarithmic axes, and so not a pure power law, and report that a model based on vascular-system architecture fails their test. The $d/(d+1)$ value of 0.750 for $d = 3$ matches the upper end of this range; it is not an empirical consensus.

12.2 Comparisons across domains

Different transport and metabolic models can yield related formulae under different assumptions (Section 2). That motivates comparison of their mechanisms. It supplies no proof that composition, space filling and conservation alone select an exponent. The conditional transport calculation makes the extra premises explicit.

The Friedmann change of variables and the cited analytic physics values are not independent measurements of this model. Unsourced fragmentation values remain excluded. A network dimension must be assigned from independent geometry or construction before it is used to predict the response.

12.3 Other reported programme results

Paper VII contains the programme's historical composition-classification comparisons. Their outcome depends on the classified observable, candidate set and scoring procedure; it is not proof that every physical process is a homomorphism. Its dated report and the associated registrations are the sources for those tests, which were not rerun for this revision.

The alignment comparison in Paper IV.d, version 2.7, reports that two earlier unblinded rises did not hold in a later run that changed several protocol components. The October code review found defects, so the comparison establishes no single cause and no correct direction. The separate Eden Protocol pilot and

Paper V report changes under scoring that was not blind, without a complete usefulness measure. These results retain their own scope; none validates a physical transport exponent. The detailed alignment and pilot accounts, including the failed run and scoring limitations, are retained in Appendix H.

Unit 83's registered performance contrast is stated with its limitations in Section 16. It is a test of correction placement on an administered task, not of the transport theorem, the Cauchy identities or a numerical capability ceiling.

12.4 Reported error-scaling measurements from the companion programme

The compute scaling question has been examined in Paper II, version 2.19, across 5 frontier architectures using 18 AIME/Putnam-level problems. Its equation (1) models the error proportion as $E(R) = E_0 R^{-\alpha_E}$; this section writes α_E to distinguish that error-decay estimand from a general capability exponent. For the reported Gemini fit, the predictor is mean generated tokens at three non-zero-error depth levels; the parallel predictor is sample count. The reciprocal-error ratio $E_0/E(R)$ shares the fitted power only under that declared model and where $E(R) > 0$. Accuracy, general capability, β_L, β_C, γ and the depth-to-time parameter δ are not thereby measured. This paper predicts no numerical value for the error exponent. Table 6 retains the reported estimates and their reliability assessments.

Table 6. Level 7: reported error-decay exponents (Paper II, version 2.19, Table 8).

Model	$\alpha_{E,\text{seq}}$	Bootstrap 95% CI	r^2	Reliability
Gemini 3 Flash	0.59 (endpoint); 0.49 (regression, standard error 0.20)	[−1.3, 2.9] (endpoint)	0.86 (regression)	**Cleanest measurement.** Monotonic accuracy curve across depth levels. Parallel exponent 0.31 (endpoint; 0.34 by regression), the arms not matched for compute. One model's fit and inconclusive.
Groq Qwen3-32B	N/A	N/A	N/A	Floor effect (about 50%, erratic).
GPT-5.4	N/A	N/A	N/A	Step function (53.7% → 100%). Not a clean power law.
DeepSeek R1	3.05	[−6.6, 23.5]	N/A	**Unreliable.** Near ceiling (94.4% to 100%); a very wide interval.
Grok 4.1 Fast	−6.62	[−58, 48]	N/A	Near ceiling (96.3% to 100%); 100% at every sample count in the parallel arm. Meaningless noise from a near-perfect ceiling.

The rows follow Table 8 of Paper II, version 2.19, on its 18 tier-2 problems; DeepSeek R1 is the model called through the identifier `deepseek-reasoner`. GPT-5.4 and Groq Qwen3 have no tier-2 sequential estimate there; their result files record endpoint estimates over all 30 problems of 1.47 (regression 1.60, $r^2 = 0.95$) and 0.24 (regression 0.09, $r^2 = 0.10$), and 0.34 on Groq Qwen3's 18 tier-2 problems, which Paper II does not interpret, because GPT-5.4's rests on a step from no reasoning to any and Groq Qwen3's accuracy shows no trend. The rows this paper recorded in March 2026, which differed, are described in the scope note of 3 October 2026 (Appendix A.3).

Interpretation. Only Gemini 3 Flash supplies an interpretable error-decay fit in this comparison. For that model, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of −1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a parallel exponent of 0.31 (endpoint; 0.34 by regression), its arms not matched

for compute. It is one model's fit and inconclusive. The two sequential point estimates are below one, and the endpoint estimate's interval extends above one. The models' order in raw performance (DeepSeek > GPT > Gemini > Groq) bears on no prediction of this paper, and most individual measurements are unreliable due to ceiling effects. The apparently super-linear exponents for DeepSeek R1 ($\alpha = 3.05$) and, over all 30 problems, GPT-5.4 ($\alpha = 1.47$) are artefacts: the first of accuracy near the ceiling, the second of a step from no reasoning to any. They are not evidence of a super-linear capability trajectory or a reliable super-linear error-suppression relation. Measuring the full cross-architecture hierarchy requires harder problem sets (IMO-level or research-grade) that keep all models away from ceiling.

Selection disclosure, before the number is used elsewhere. One clean fit in five is not by itself evidence of the framework. The four traces that could not be fitted are not neutral absences: they are outcomes of the same experimental design that produced the fifth, and Gemini 3 Flash is reported first because it was the only model whose scaling regime was clean enough to fit a power law at all, a selection made after the results were seen. On its own, this sweep cannot discriminate the framework from a null in which one clean sub-linear curve arises by sampling; the bootstrap interval $[-1.3, 2.9]$ is wide even within that selected fit and does not account for the post hoc selection of the model; the harder-problem re-run described above, keeping every model off the ceiling, is the route prepared as a draft registration awaiting human submission, toward a discriminating estimate. The 0.49 value, the regression exponent, is therefore always quoted with the rest of its row and the fixed-weight settings tested, as the interpretation above quotes it, never as a stand-alone confirmation.

The parallel sampling finding. The five runs' result files record parallel exponents of about 0.00 (DeepSeek R1), -0.03 (GPT-5.4), 0.31 (Gemini 3 Flash, by endpoint estimate; 0.34 by regression), 0.22 (Groq Qwen3, over all 30 problems) and none for Grok 4.1 Fast, at 100 per cent at every sample count, so the parallel exponent was near zero in two of the five; the reported error-decay exponent was above the parallel one wherever both could be estimated, in four of the five runs, and the two arms were not matched for compute.

13 Equations and what each requires

Table 7. Different equations describe different quantities.

Relation	Premises and interpretation
$C = B\bar{\ell}_q$	Steady nonnegative routed flow; C is flow-weighted path cost, not automatically mass or energy
$\alpha_C = \mu/(\mu + \zeta)$	Demand and mean-path asymptotic scalings; $d/(d+1)$ additionally sets $\mu = d$ and $\zeta = 1$
$N = C/v$	Uniform controlled transport speed; varying speed changes the inventory exponent
$h = b - s - e$	Steady oxygen balance; heart-rate scaling depends on metabolic, stroke-volume and extraction exponents
$d_{\text{eff}} = 2/(1 + 3w)$	A coordinate change in the stated flat cosmological solution, not measured network geometry
$\alpha = 1/(1 - \beta_L)$	Asymptotic exponent of the declared gain ODE for $\beta_L < 1$; finite-depth elasticity need not equal it

A common numerical fraction does not identify a common mechanism. The operational quantities, domains and assumptions in the preceding sections govern every use of these equations.

14 Predictions and a prospective transport test

The earlier predictions retain their numbers. Their current dispositions below correct this paper's interpretation; they alter no frozen registration, data or registered verdict.

Prediction 1 (The earlier one-dimensional organism prediction)**.** The earlier paper predicted a metabolic exponent of one half for organisms with one-dimensional internal transport. That number follows from the present transport model only when the demand, route, speed and mass premises are established. The fungal measurements do not yet establish them: two of the selected fits use colony respiration against biomass, while the saprotrophic fit uses specific metabolic rate against radial extension. Their separate variable pairs and uncertainties are retained in Appendix F; the former pooled confirmation is withdrawn. A valid test needs independently established mechanism variables and a declared response, rather than classification from the observed slope.

Prediction 2 (withdrawal recorded in version 3.13, 30 October 2026)**.** Withdrawn. The published record refutes it.

Before the version 3.13 manuscript, this paper predicted that the neural scaling exponent (of loss against parameters) would equal $d/(d+1)$, with d the intrinsic dimension of the training data's manifold. Once d reaches 1, that fraction is never less than one half, and it climbs towards 1 as d grows. The exponents measured for language models lie far below it. For language models trained to convergence on large datasets, Kaplan et al. (2020) report about 0.076 (their equation 1.1). For the parameter term of the loss, with its irreducible part set apart, Hoffmann et al. (2022) report 0.34 (their equation 10). They note that both of their exponents are lower than one half. A refit of data reconstructed in part from one of their figures gives 0.35 (Besiroglu et al. 2024). To reach such values the formula would need data of less than one dimension, about 0.08 and 0.5. The intrinsic dimensions estimated for language and image data run from about 7 to more than 90 (Sharma and Kaplan 2020; Pope et al. 2021).

The dependence runs the wrong way too. In a transport network a higher dimension lifts the exponent towards 1. In neural networks the measurements run the other way. Where the dimension of the data is dialled in controlled teacher and student experiments, the measured exponent falls, close to $4/d$ (Sharma and Kaplan 2020; Bahri et al. 2021), while $d/(d+1)$ rises. The account given for this is geometric (Sharma and Kaplan 2020; Bahri et al. 2021). On that account, a network divides its data manifold into pieces, and halving their size takes 2^d times as many parameters. So the higher the dimension, the more slowly the loss falls. The two curves cross once, near $d = 4.8$, so a single dataset of about that dimension could not tell them apart.

The link between this exponent and the dimension of the data manifold is earlier work's. Sharma and Kaplan (2020) proposed that the power law can be explained if networks are, in effect, performing regression on a data manifold of intrinsic dimension d. For networks built from rectified linear units, with cross-entropy, mean-squared-error or KL-divergence loss and plentiful data, they predicted $\alpha \approx 4/d$. They stated it as a bound, $\alpha \gtrsim 4/d$, with equality expected in the generic case. They confirmed it by measuring the dimension and the exponent independently in teacher and student networks, and tested it on image classifiers and GPT-type language models. For the language models they found d above 90 against $4/\alpha$ of about 53, in accord with the bound but, in their words, quite far from equality. Bahri et al. (2021) identified four regimes: variance-limited and resolution-limited, each for dataset size and for model size. In the variance-limited regimes, for smooth test losses, the loss approaches its limit with exponent 1, in dataset size or in network width, or in parameter count for linear models. That exponent is independent of the architecture and the data. For the resolution-limited regimes they argued that the exponent is inversely proportional to d, and found that teacher-student exponents closely match $4/d$. In the large-width limit they derived these exponents from the spectra of kernels, and presented evidence that the model-size and dataset-size exponents are related by a duality. Earlier still, Kaplan et al. (2020) pointed to power-law

scalings in density estimation and random forests. Such models, they wrote, suggest that "power-law exponents may have a very rough interpretation as the inverse of the number of relevant features in the data".

The measurements that refute the prediction were published between 2020 and 2022, years before it was printed. It is withdrawn, not adjusted. The withdrawal does not decide the separate conditional transport model of Section 6. Its physical applicability requires the premises stated there.

Prediction 3 (Prospective composition classification)**.** The earlier criterion compares a family assigned before fitting with the later response. A classical theorem cannot be empirically contradicted while all of its exact premises hold. The scientific test is whether the independently assigned composition model and its measurement assumptions predict held-out observations. The classification rule, alternatives and uncertainty must be fixed before those observations are examined.

Prediction 4 (The earlier geometric-bound prediction)**.** The original claim barred exponents at least one for any finite-dimensional hierarchical network with multiplicative composition. Those descriptions alone do not prove that bound. Its universal reading is withdrawn as a consequence of the argument. The narrower conditional throughput-cost bound in Section 10 states the extra premises; adopting it in an experiment requires a new prospective protocol and cannot relabel an earlier result.

Prediction 5 (The separate Eden Protocol hypothesis)**.** The earlier behavioural hypothesis concerns whether explicit purpose, stakeholder-care and universalisability checks improve alignment-relevant output quality, with usefulness measured alongside it. It is not derived from network dimension. The reported pilot used scoring that was not blind and measured no complete usefulness outcome. Its numerical record and the later suite remain in the historical account and Paper V. A protocol comparison cannot by itself identify internal ethical reasoning as its cause. Any replication must separate the intervention, the evaluation procedure, task completion, needless refusal and resource cost.

14.1 The next discriminating measurement

The proposed question is whether independently measured demand and route scaling predict throughput against resource inventory on held-out networks. It is an unrun design proposal. No apparatus, supplier quote or adequately powered sample has been established here, and it adds no commitment to an existing funded project.

1. Define an eligible transport apparatus or dataset with independent measurements of throughput, inventory, size, route lengths and speed. State the observational unit and exclusions. Do not manufacture an inventory observation from the accounting equation being evaluated.
2. Fix the size range and one routing intervention. For a declared increase of every served route length by the same factor $c > 1$, with demands and speeds unchanged and no extra waiting, inventory should increase by the factor c at each size; the exponent need not change. A claim about a changed exponent instead needs an intervention that changes the size dependence of routes or another specified scaling premise. State which comparison is being tested, and establish geometry before fitting the response.
3. Split complete independent networks into calibration and held-out sets. Use calibration records to specify the finite-range functions for demand, route length and speed, including any offset, curvature or finite-size correction. Lock their joint uncertainty and the predicted held-out trajectory. An exponent-only prediction is admissible only if the protocol justifies a power approximation and a bound on its error over the actual size range (Equation (18)).

4. Measure held-out throughput and inventory independently of their predicted values. With the declared steady-flow premises, the finite-range prediction is $N(L) = B(L)\bar{\ell}_q(L)/v(L)$, with all three right-hand functions fixed from calibration. Model the measurement errors and any shared calibration covariance. Under justified power behaviour and uniform fixed speed, the limiting target exponent is $\mu/(\mu + \zeta)$. For a separately specified uniform-speed scaling $v \propto L^{\upsilon}$, it becomes $\mu/(\mu + \zeta - \upsilon)$ when the denominator is positive. Do not fit a missing mechanism variable after seeing a discrepancy.
5. Compare the predicted model with a free-exponent power law and a justified curved alternative on the same response and likelihood. Prespecify measurement-error treatment, uncertainty, exclusions and multiplicity. The choice between a log-transformed fit and nonlinear fitting must follow the assumed error distribution, rather than a universal preference for one method (Xiao et al. 2011). Use design analysis to establish sample size and a meaningful discrepancy tolerance before validation; inadequate precision returns the declared inconclusive outcome, not a post-hoc relabelling of the study.
6. Publish the prediction, observations, deviations and all planned comparisons. A premise that fails is itself a limitation to report. A precise predictive failure cannot be repaired by changing dimension, the outcome or the model after inspection.

The accounting identity is mathematical. The empirical questions are whether its non-definitional premises and measurement bridges apply, whether independently estimated parameters transfer, and whether the prediction beats serious alternatives. Agreement would support that specified mechanism over the tested range. It would establish no universal metabolic law or AI alignment guarantee.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. The earlier dated predictions remain available beside the corrected argument.

15 Making the composition claim measurable

The useful chain is from a specified observable and mechanism to a locked prediction. A family label alone does not supply an exponent; a repeated numerical fraction does not establish a common mechanism. The homomorphism result, the transport accounting and the capability ODE answer separate conditional questions. This section sets out what a test must supply to connect them to observations.

1. Declare what is measured. Specify the response, input, units and both composition operations.

⇓ *Classical mathematics constrains a family, given its premises.*

2. Specify the physical model. Measure demand, routes and speed; justify the inventory or mass bridge.

⇓ *The transport assumptions select an exponent for a particular variable pair.*

3. Test a locked prediction. Use held-out systems and independently measured outcomes, against prespecified alternatives.

Figure 1. From a conditional mathematical statement to an empirical claim. A theorem does not supply the physical premises; calibration and validation must not recover the same constructed identity. No observed data are plotted.

Definition 4 (A measured baseline)**.** For a positive response U and a declared reference input R_0, define the baseline and normalised response by

$$I = U(R_0) > 0, \qquad G(R) = \frac{U(R)}{I}, \qquad G(R_0) = 1. \tag{47}$$

The response, its units, the reference condition and the population to which it applies are part of the definition. $R_0 = 1$ denotes a first-pass baseline when that convention is appropriate. If the baseline is zero, this ratio is undefined and a different, prospectively specified measurement is needed.

This baseline is not automatically entropy reduction in bits, intelligence or self-correction capacity. Accuracy, logical error, metabolic rate and phase coherence are different observables. A common symbol cannot supply a conversion between them. Information-theoretic quantities require a specified random variable and probability distribution; energy converted into useful performance requires a physical and measurement model. Landauer's lower bound on irreversible erasure does not make every energy flow a measure of information gained (Landauer 1961).

15.1 The assumptions to fix before validation

1. **Observable and axis.** Name the response, units, baseline, admissible values and input. Distinguish additive passes, multiplicative scale ratios, code distance, qubit count, organism mass and elapsed time. State the transformation used before interpreting a slope.
2. **Both operations.** State the input operation and output operation and why the observable is expected to preserve them. Separate that equation from the state-update rule. If the operation is Equation (45), test that operation rather than silently replacing it by ordinary multiplication of depth.
3. **Exact, approximate or asymptotic claim.** Name which is proposed. For an approximate identity give the residual norm, domain and error bound. For a scaling limit state the ratio, limiting variable and possible slowly varying factor. Finite observations test a specified finite-range consequence; they do not verify every point of a global hypothesis.
4. **Exponent-selecting assumptions.** In the transport model, measure demand scaling μ and flow-weighted path scaling ζ; justify a demand-based assignment $\mu = d$ and any bridge from transport to mass. In the gain ODE, identify a and β_L independently and predict the complete shifted trajectory. Neither dimension nor coupling may be inferred from the validation exponent and then presented as its prediction.
5. **Resource and clock controls.** Identify which weights, code, memory, prompts, tools and evaluation procedures change. Record hardware, total computation, latency and external help. A change in resources or cycle duration must not be mistaken for the claimed improvement mechanism.
6. **A discriminating outcome.** State which observation would favour or reject this conditional explanation against realistic alternatives. Specify the uncertainty, the error model, dependence between observations, saturation and the precision needed for an inconclusive result. An intervention on routing, information access or the improvement rule is stronger evidence of mechanism than another fit to the same curve.

15.2 An independent prediction needs separate information

Assign complete systems or runs to calibration and validation before inspecting outcomes. Use calibration to justify the mechanism and estimate its parameters. Lock the predicted trajectory, uncertainty, variable definitions, exclusions, stopping rule and decision margins before validation. Compare exact and shifted powers, exponential, logarithmic and appropriate bounded or broken-form rivals on the same response with compatible likelihoods. A repeated observation from one system is not an independent system, and a parameter obtained by transforming the fitted validation slope is not an independent prediction.

For the network question, independently measured demand and route scaling should predict later throughput against transport quantity; a routing intervention has a specified effect on the whole predicted curve. Multiplying route lengths by a size-independent factor while holding demand and speed fixed

changes inventory by that factor, but need not change an exponent. An exponent-changing intervention must instead alter a scaling premise. Other declared controls remain fixed. For a capability question, the validation task must distinguish improved task performance from an improved ability to produce subsequent improvements. These are design requirements, not new results or a registration made by this paper.

Report departures from the premises and failed predictions rather than moving a system to another class after seeing the answer. A useful diagnostic can motivate a new prospective test; it cannot turn the original failure into success. Analytic identities, retrospective fits and prospective results retain their different evidential roles. The existing withdrawal of Prediction 2 and every frozen registration remain unchanged. The programme's laws remain laws under test, with verdicts taken only through their registered rules. No phase diagram of all complexity, universal physical exponent limit or cosmological ladder is needed to carry out this measurement programme.

16 What this paper does and does not show

The paper establishes conditional mathematical statements and a prospective way to test their applicability. The four chosen homomorphism problems yield classical function families. The transport calculation yields an exponent only under explicit demand and path scalings; mass and inventory require further bridges. The numerical resemblance of exponents across fields supplies no additional proof.

Its biological comparisons are source-specific, its cosmological map is analytic, and its AI summaries retain the limitations of their own reports. The former pooled biological inferences, universal dimension-only bound and physical complexity ladder are superseded. Prediction 2 remains withdrawn. No new observation or registered verdict is reported in this revision.

Status as at this version. The theory registration was registered on 8 September 2026 at version 1.100 and updated on 13 September (1.102), on 15 October (1.103, which registered Laws IV and V) and on 29 October 2026 (1.104); the earlier texts stay frozen beside the current one. What this programme names laws are conjectures under test. Laws I to III were registered on 8 September 2026 and Laws IV and V on 15 October 2026 (version 1.103). Every law is a law under test until a registered test returns its verdict. The mathematics of Laws IV and V is still being developed (version 1.2, 29 October 2026). Law IV is not yet tested. Law V is a law under test: its placement limb, P21, is supported on the performance axis only, under the registered test and its refusal score, by the programme's own run (Unit 83, 28 October 2026, DOI 10.17605/OSF.IO/MN73W; results: DOI 10.17605/OSF.IO/98HKR): family-dependent, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism, and one disclosed recovery amendment; its build-order limb, read on P12, is undecided, since P12 is untested. The first registered study, Unit 06 (a classification audit, DOI 10.17605/OSF.IO/RV2B5), was registered on 1 October 2026 and run on 6 October 2026; it is reported in full with its five deviations as administratively incomplete, with no inferential verdict (results DOI 10.17605/OSF.IO/4BPHG), and it tests no law. Unit 82 was registered on OSF on 27 October 2026 (DOI 10.17605/OSF.IO/JUFG4) and has not yet run. No law is confirmed. The new transport quantities have explicit units and definitions. The gain ODE defines β_L mathematically, but a valid independent estimator for any target AI system still needs to be specified and tested. Established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

17 Limitations

- The exact composition identities and global stability assumptions require independent justification. A finite noisy fit cannot establish them across an unobserved domain.
- The transport exponent is asymptotic; finite-range prediction requires a full curve or a justified bound on finite-size corrections. It depends on demand, paths, speed and the chosen response. A measured spatial dimension alone supplies none of those extra assumptions. Body mass needs a physiological bridge.
- The biological estimates concern different taxa, conditions and variables. The fungal factors cannot be pooled into one metabolic relation. Selected published exponents do not establish a universal law.
- The cosmological reparametrisation has no independent mechanistic content. Photon trajectories and a cosmic web are not assumptions of the homogeneous background solution.
- Earlier computational and behavioural results retain their design, selection, scoring and amendment limitations. This revision reruns none of them and claims no new confirmation.
- The prospective transport test still needs an eligible apparatus or dataset, practical resource estimates, independent replication units and a design-sensitivity analysis before registration.
- The synthesis relies on established mathematical and transport literature. Its value must be demonstrated through a useful distinction or successful prospective prediction; a larger catalogue of matching fractions would not establish that contribution.

18 Conclusion

A useful scaling prediction needs more than a familiar curve. It needs declared variables, a justified composition or transport model, and a test on observations that did not select the assumptions. The classical equations constrain families; the conditional transport model shows where an exponent enters. Neither result classifies all of nature.

The corrected biological and cosmological examples show why these distinctions matter. Heterogeneous measured slopes cannot supply one pooled validation, and an algebraic change of variables cannot supply a physical mechanism. Fixed computational rules can generate superlinear trajectories, while correction and safety require additional quantities and models.

The next empirical contribution is the prospective transport comparison: estimate the mechanism separately, seal the prediction and allow the held-out observations to support it, contradict it or remain inconclusive. The earlier record stays visible. The purpose of the revision is to make the surviving argument precise enough to be tested and useful enough for another researcher to challenge.

Code and data. No study or data analysis was run for this revision. The historical suite `arc_complete_test_suite.py` and its original results remain byte for byte in the dated public repository. It implements earlier computations, including the pooled and three-group analyses withdrawn here; it is not an implementation or validation of the new prospective protocol. The paper's existing deposit identity and earlier editions are retained. Source transformations and checks document this revision separately; they are not experimental outcomes.

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Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Version history and corrections

This appendix separates the present source revision from the earlier dated record. A correction is dated when it is made; retaining an earlier passage does not make its present correction part of the earlier claim. Table 8 records the changes in this review. The subsequent history preserves what the earlier editions said, including their claims and the limits of their corrections.

A.1 Version 3.14: revised 1 November 2026

Status: candidate awaiting final review. The source review was prepared on 30 October 2026; the present candidate revision is dated 1 November 2026. It does not record a public deposit, a registration, a study run or a new empirical result. The earlier version remains intact. Its full content is retained through the current argument, the source-measurement and companion records, and the explicitly historical material; superseded interpretations are not silently recast as earlier statements of the new conditional argument.

The outside review considered version 3.12. Several of its principal corrections were already present in version 3.13: the restricted abstract, the distinction between metabolic scaling and capability against depth, the Friedmann boundary calculation, the withdrawal of mixed error aggregates, and the withdrawal of Prediction 2. The table below distinguishes those retained corrections from the further changes made in this revision.

Table 8. Corrections recorded for the source review of 30 October 2026.

Subject	Current correction and its scope	Current location
Composition and its domain	The four chosen homomorphism problems now have explicit source and target groups: the additive reals and the multiplicative positive reals. The logarithmic output is real-valued. Normalisation, constant cases and the rational-to-real proof are stated. This repairs the domain statement; the classification is not extended to all physical laws.	Section 4
Approximate and asymptotic composition	The stability statement now gives a global logarithmic defect bound and a proof of the resulting multiplicative band around a power. Finite observations do not establish that global premise. A separate regular-variation result is credited to its established literature. The earlier withdrawal of dynamical-attractor language is retained, not dated anew.	Sections 2 and 4
Transport exponent	The current result starts from the flow-accounting identity $C = B\bar{\ell}_q$. Demand and mean path scaling give the conditional exponent $\mu/(\mu + \zeta)$; $d/(d + 1)$ follows only from the stated special choices. This replaces the unproved claim that dimension, space filling and conservation alone select the exponent.	Section 6
Inventory, mass and the bound	Flow-weighted transport cost, inventory in transit and body mass are distinct quantities. Speed and the mass bridge require their own premises. The geometric bound now states the demand-distance and scaling assumptions for its throughput-cost comparison. It is not a bound on every physical observable or on capability against recursive depth.	Sections 6 and 10
Heart rate	The Fick oxygen balance makes metabolic, stroke-volume and oxygen-extraction assumptions explicit. The heart-rate exponent is their difference, not a consequence of dimension alone. Behavioural state and prefactors remain relevant. The earlier correction of unequal lifetime beat totals is preserved.	Section 7

Subject	Current correction and its scope	Current location
Biological comparisons	Each fungal measurement keeps its own variable pair and uncertainty. Two biomass-related factors and a radial-extension-related factor no longer serve as a pooled validation of one metabolic slope. The disowned three-group comparison and mixed aggregates are retained as historical analyses, not current validation. Source-specific measurements and the competing metabolic evidence remain available.	Sections 12 and 14
Cross-domain comparisons	An exact exponent, an algebraic reparametrisation, a measured slope and an unsourced value retain different evidential roles. A shared fraction does not establish a transport mechanism. Earlier physics and biological catalogue rows are preserved with their sources and corrections; they do not supply a new aggregate accuracy or a universal validation claim.	Sections 6 and 12
Cosmology	Version 3.13 already corrected the boundary: $w = -1/3$ gives linear expansion, $-1 < w < -1/3$ accelerating power laws, and $w = -1$ exponential expansion in the specified flat, single-fluid model. That correction remains. The photon-path and cosmic-web mechanisms and the dimensional cosmic ladder no longer stand as explanations of these homogeneous solutions. The mapped dimension is an algebraic coordinate.	Section 9
Recursive gain and phase claims	A fixed-rule quadratic recurrence separates iteration from sublinearity. The assumed gain equation is given with its shifted solution, finite-depth elasticity, parameter domain and deformed composition operation. A superlinear power is not an exponential phase. Intelligence-as-phase-transition and universal complexity-phase claims remain historical interpretations rather than current conclusions.	Sections 11 and 15
Correction axes and measured quantities	Code distance, physical-qubit count and elapsed correction cycles are separated, as are per-cycle suppression and cumulative failure. The two 2024 source dates remain. The current baseline I is declared in the response units; the former universal entropy-reduction definition is retained historically. Landauer's erasure bound is not an energy-to-capability conversion.	Sections 11 and 15
Predictions and programme results	The five prediction numbers and their dated record remain. Prediction 2 was withdrawn in version 3.13; its source-based reasons remain current. The other dispositions state the premises now required. The new transport protocol is prospective and unrun. No frozen registration, study data or registered verdict is changed by this source revision, and no new empirical result is reported.	Sections 14 and 16
Contribution and priority	The contribution is a conditional mathematical and methodological synthesis, with prior models compared at their actual scope. Regular variation and the prior homomorphism connection are credited, including Bingham et al. (1987) and Ostaszewski (2016). Earlier broad novelty claims remain dated evidence of what was written; they are not current claims of a new universal theorem or a first connection.	Section 2; Appendix E
Dates and preservation	The recorded first-publication date remains 22 February 2026, with its stated documentary basis. The 16 August 2026 working-paper header line remains a separate historical record whose significance has not been established. Neither is reassigned by this review. The earlier source, dated corrections, measurements and priority records are preserved; the present corrections are dated 30 October 2026.	Appendix B

Reconciliation with the released version 3.13. The source review above began from the version 3.13 manuscript prepared on 30 October 2026. Version 3.13 was subsequently released with the edition date

31 October 2026. Its release pass retained the Prediction 2 withdrawal, updated the cited Laws IV and V mathematics to version 1.2, named registration 1.104 where current registration was intended, and added the separate Unit 83 results reference (Eastwood 2026ac). It also replaced fungal agreement language with "not rejected, no equivalence shown", while retaining the published values and identifying the different measurement axes. The present version goes further: those unlike factors do not support a pooled metabolic inference. That correction is recorded in the 30 October source review, rather than retrospectively attributed to the released version 3.13.

The 1 November continuation retains the earlier dated source record below, reconciles the later release changes, dates this candidate separately from first publication, restores the approved contents-before-introduction presentation and uses the already proved local bibliography style whose year suffixes continue aa and ab after z. These presentation and provenance changes add no empirical result and do not change the conditional mathematical amendments already recorded above.

Further source corrections, 1 November 2026. The companion table now identifies Paper II's error-decay exponent from its equation (1), rather than calling it a general capability exponent. Every reported estimate and uncertainty is retained. It supplies no numerical estimate of the capability-gain, correction-capacity or cycle-time parameters. Appendix H restores the detailed alignment and pilot reports, the failed run, the earlier heartbeat illustration, the superseded pooled and phase interpretations, the code provenance and all three predecessor figures that were preserved in source records but missing from the initial review PDF. Each is labelled with its date and scope; no withdrawn inference is restored as a current finding.

How to read the retained history below. The following record is carried forward from the version 3.13 manuscript prepared on 30 October 2026, before its release pass of 31 October. The preceding paragraph records the later release changes. Its wording is historical: "this version", "now", result grades, novelty statements and registration status refer to the edition or date named there, not to the present review. References inside that retained record use the section, equation, table, figure and appendix numbers printed in version 3.13, unless a passage itself names an earlier edition. The original reference numbers are fixed here so that a historical pointer cannot silently lead to a changed current equation or diagram. In particular, Figures 1, 2 and 3 below are figures of version 3.13, not current figures. Table 9 retains the mapping from version 3.12 to version 3.13; it is not a map into version 3.14. Current claims and status are given in the main text and Section 16.

A.2 Version 3.13

v3.13, 30 October 2026: Substantive. Prediction 2 is withdrawn: the published exponents of loss against parameters for language models, and the measured fall of the exponent as the data's dimension rises, refute it. Section 14 now says so, gives the measurements and credits the earlier work on the exponent and the dimension of the data manifold (Sharma and Kaplan 2020; Bahri et al. 2021). Level 6 of Section 12 no longer calls the prediction untested, and a dated correction note records what both passages printed before. The works of Sharma and Kaplan (2020), Bahri et al. (2021), Hoffmann et al. (2022), Besiroglu et al. (2024) and Pope et al. (2021) are added and Kaplan et al. (2020) is now cited. No registered law, registered prediction, kill condition or dependency changes; no other result is added or changed by the withdrawal; no date of any claim moves.

This version is also the paper's edition on the programme's presentation standard, typeset in LaTeX. The sections of version 3.12 keep their order, except that Section 12, What Is New, becomes Section 2; Table 16 maps every earlier section to its place here. The title now names the paper's domain, recursive composition, as the scope correction of 22 August 2026 requires; until this version it read "Framework

Paper · The ARC Theory · Law I, the ARC Principle · On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind the Law". The subtitle "A Universal Principle of Recursive Composition Across Physics, Biology, and Cosmology", which said what the scope correction withdrew, leaves the title block. The abstract is rewritten within arXiv's limit, and the abstract of version 3.12, with its scope correction, is kept in Appendix A.2. The first-page footnote gives the paper's place in the programme, its version, its DOI, links to the author's page and his book, and its licence; the closing note on authorship becomes the acknowledgement, unchanged.

The corrections owed to this version are made, each with its source:

- Laws IV and V are stated as registered as laws on 15 October 2026 in version 1.103, Law IV not yet tested and Law V a law under test, their mathematics at version 1.1; the epistemic-status line gives the registration of all five laws and says that every law is a law under test until a registered test returns its verdict, and Section 16 gives Law V's status with the registered result of Unit 83, whose results are public (DOI 10.17605/OSF.IO/98HKR).
- The March 2026 alignment run is described as one unregistered run of six models from six providers, its design named as the design, its 2,170 responses each scored by five or six distinct judges, its tiers given under the October 2026 code-review caveat, and the change between the earlier experiment and that run as Paper IV.d (version 2.7) states it (Section 12).
- The parallel exponents are given as the five result files record them, the two arms not matched for compute, and GPT-5.4's starting accuracy as 53.7 per cent; the parallel finding is no longer called the most robust result (Section 12).
- No result from the analysis of 11 August 2026 is printed; the 19 of 25 is given under the classification rule of March 2026 (Section 12 and Appendix D).
- Cost estimates are no longer printed (Section 14).
- The Eden Protocol's loops are named by what they do, with the book's names given once (Section 14).
- No form of "propose" is used for the author's own work; the ARC Bound is named as the conjectured value of Law III, the ARC Ceiling, a law under test; Paper II's replication is named as it is titled; Unit 82, registered on 27 October 2026, is named where the paper said the value two was on trial in drafted registrations (Section 11).
- The physical and cosmological claims are set out at their true status: the bound is scoped to systems whose growth is carried through a space-filling hierarchical network, with the counterexamples conceded (Section 10); intelligence is no longer called the one exception to it, the difference being the pair of variables (Section 11); $E = mc^2$ is stated as the energy of a mass at rest (Section 8); the cosmology is stated as an identity and its causal reading as this paper's interpretation, beside what established cosmology says (Section 9); the error-correction regimes follow the published threshold result (Section 11); Landauer's principle is stated exactly and the definition of I is graded as a working definition (Section 15); the space of laws is stated as finitely many forms in a finite-dimensional family (Sections 2 and 15); the comparison with Einstein's work keeps its concession without the self-comparison (Section 2); the closing quotation is given in the 1936 article's wording (Section 18).
- The dated notes of 19 September 2026 and 3 October 2026 are carried into the passages they govern and kept whole in Appendix A.2; the subheading "Why there cannot be a fourth" now reads "Why the grid has four cells and no fifth".
- The status of $d = 2$ in biology is carried through: the three-group comparison, computed with a group the paper sets aside, is qualified; the filled marker at $d = 2$ in Figure 3 is no longer drawn; the legend of Figure 2 gives the paper's own grades; Prediction 1's closing sentence no longer claims a

confirmation at $d = 2$ or that no theory has achieved one.

- The fungal value 0.547 is given as the mean of the three colony-level scaling factors that Aguilar-Trigueros et al. (2017) report at 20 degrees Celsius, where the biology-cosmology mirror printed it with a standard error of 0.07 and the dimensional ladder drew that bar; 0.07 is the standard error of one of the three, not of the mean. The p-values beside it are named as one-sample t tests of the three, and the bar in Figure 3 now spans the three factors (Sections 9, 12 and 14).
- No count of independent derivations is given; each earlier derivation is named as its own work.
- Works are linked to their official sources, every mention of the author's papers to its OSF DOI, and ranges are written with "to".
- The boundary $w = -1/3$ is no longer called the boundary between power-law and exponential growth: under the change of variables it is where expansion turns from decelerating to accelerating and $\alpha \to 1$; past it the expansion is an accelerating power law, exponential only at $w = -1$; and in Cauchy's classification $d \to \infty$ is the linear limit of the power law (Sections 2, 9, 12, 13 and 15).
- Saturation is no longer presented as a Cauchy case, and a saturation curve is no longer called the only smooth way to approach a ceiling; Section 5 no longer says that every growing system in nature amplifies itself recursively, and it names the ARC Principle as Law I, a law under test (Sections 3 to 5, 11.1 and 15).
- Section 12 no longer says that the framework predicts a sub-linear exponent of capability against depth, or that the models' order in raw performance is consistent with the framework: the paper predicts no value for that exponent.
- Thommen et al. (2019) are reported in their own terms: they find Kleiber's three-quarter scaling in a planarian and trace it to energy storage, where the paper had said that planarians scale as surface-area theory predicts (Sections 6 and 12).
- Prediction 3's falsification criterion now requires the composition to be identified, on input and output, before the data are fitted; as printed, the paper's own logarithmic case met it (Section 14).
- The fungal t tests no longer reject the $d = 2$ value: they set aside each factor's own standard error, and each factor's own interval includes $\frac{2}{3}$ (Sections 12 and 14).
- The two brittle-fragmentation rows, printed with measured values of 0.670 and 0.750 and no source, now read not yet sourced; Figure 2 no longer draws them, and the introduction no longer states their exponents (Sections 1, 6, 9 and 16).
- The tables of the chain reaction and of quantum error correction no longer print 'ARC Prediction' and 'Confirmed' for regimes the framework only classifies, and a reactor held at $k \approx 1$ is described as running at steady power (Sections 8 and 11.1).
- The heartbeat arithmetic is corrected: the paper's own figures give about 0.6 billion beats for a mouse and about 1.0 billion for an elephant, where it said about 1.5 billion for every mammal; the heart-rate allometry is credited to Stahl (1967) (Sections 7 and 12).
- The reference to He and Chen (2003), *Fractals* 11(4), page 437, is withdrawn: no registry holds a work with that title, volume, number or page (the journal's volume 11 ends at page 417), and the paper no longer lists it among the earlier derivations (Sections 2 and 12).
- The note on the $\frac{3}{4}$ debate gives White and Seymour (2003), Savage et al. (2004) and Kolokotrones et al. (2010) what they state, no longer prints an estimate of 0.687 that no work read here gives, and no longer says that the framework predicts the variation; the abstract says that the agreement at $d = 3$ is on the group means compiled here and that the mammalian exponent is debated (Section 12).
- Each biological value in the table of predicted and measured exponents is given its source, read

at that source; the birds value is corrected from 0.720 to 0.723, the non-passerine exponent of Lasiewski and Dawson (1967), which moves the three-dimensional mean of Level 2a from 0.748 to 0.749 and its p-value from 0.858 to 0.888; the aggregate error of 2.5 per cent, which counted the two cosmology identities, is withdrawn; and Figures 2 and 3 draw the tables' values where they drew 0.746 (Sections 6 and 12).

- The percolation row states the quantity it compares, the specific-heat analogue of two-dimensional percolation, whose exponent is $+2/3$, and the KPZ row is graded as an exact value that no derivation here links to $d/(d+1)$ (Section 6.1).
- The rows of the Level 7 table follow Paper II, version 2.18 (Section 12).
- The Eden Protocol pilot is graded as supported under scoring that was not blind, where Prediction 5's status read 'partially confirmed' and the pilot was said to give the developmental hypothesis its first multi-model empirical support; the later six-model suite of Paper V is given, and the prediction names its usefulness side (Section 14).
- The universe is said to exhibit two of the three ARC patterns, as its table shows; Phase I is given about 7 billion years, not the universe's whole age; the sustained acceleration and heat death of the dark energy phase are stated with their condition, that dark energy stays constant; and intelligence is no longer said to scale without limit (Sections 9, 11 and 15).
- Section 15 states its readings as readings, among them the laws of physics as writable in the forms rather than instances of them, and the conclusion says that the universe can be described with three composition operators, where it said that it builds itself from them (Sections 15 and 18).
- Credits are stated as the works state them: Cauchy's theorem for continuous functions, the measurable and bounded forms being later results; Hyers (1941) for the additive equation, at a question of Ulam's; Shannon (1948) for the entropy of an information source; Kleiber for the 0.74 power over thirteen groups of animals; Fahrenheit (1724) for a reproducible temperature scale, not for temperature; and Einstein (1915) now has a reference entry (Sections 4, 9 and 15, and Appendix E).
- The account of convergence, and the exponent's independence of selection, are stated as conditional on the sufficiency of the three conditions, which this paper has not proved (Sections 2, 12 and 15).
- Related work adds the standard result that invariance under rescaling forces a power law (Newman 2005) and the critiques of West, Brown and Enquist's derivation (Dodds et al. 2001; Kozłowski and Konarzewski 2004; Savage et al. 2008), and says what He and Zhang (2004) and Maino et al. (2014) derive (Section 2).
- The why in the abstract and the introduction speaks of the family of forms that growth can take, since the exponent of capability against depth has to be measured.
- The introduction's heart rates are given with their sources, the blue whale's as the 4 to 8 beats a minute of a foraging dive; its fungal exponent is given as colony-level measurements show it; and a city ten times larger needs about six times more petrol stations, not about seven (Bettencourt et al. 2007) (Sections 1, 3 and 7).
- The alignment run of March 2026 is dated from its own evidence, its tested predictions public before it ran, and the combined finding gives its tiers as three positive, two flat and one negative, where it said that alignment does not scale for most architectures (Section 12).
- The quantities the paper compares with data are said to be operationally defined, not every quantity (Section 16); Equation (14) and Law III's ceiling are said to share a form, and the two meanings of β are listed (Section 11 and Appendix C); and the plain-English reading of $p = 0.003$ counts a change in either direction (Section 14).

- The saprotrophic rows are credited to Crowther and Bradford (2013), not to Wilson and Griffin (1975), and their third temperature is 28 degrees Celsius, not 25; the reference to Hartikainen and colleagues (2014) gives the published work's title, issue and pages; He and Zhang's second author is Juan Zhang; and Aguilar-Trigueros et al. (2017) are said to have compiled published measurements in what they believe is the first such synthesis (Sections 12 and 14).
- Headings and display labels name what the sections and equations do, among them Section 8, Section 11.1 and the labels of Equations (8), (13) and (14); Levels 2b to 6 of Section 12 share one subsection; the slogans move to Appendix D.4, the analogy with Einstein's work to Appendix E, and the withdrawn box of Section 8 and the closing line of version 3.12 to Appendix A.2; and the Code and data paragraph says what the published suite re-runs and links the commit it names.

A.3 Earlier corrections and notes

Correction note, 30 October 2026. Section 14 printed Prediction 2 as: "The neural scaling law exponent (loss vs parameters in machine learning) should equal $d/(d+1)$ where d is the intrinsic dimension of the training data manifold." Level 6 of Section 12 called it untested; the published record had already tested it. The exponents of loss against parameters published for language models from 2020 lie below one half, the least value $d/(d+1)$ takes once d reaches 1, and in controlled experiments the measured exponent falls as the data's dimension rises, where $d/(d+1)$ rises. The prediction is withdrawn. The earlier work on this exponent and the dimension of the data manifold is credited where the prediction stood, and Kaplan et al. (2020), listed until now without a citation, is cited there. No other result of this paper changes with it, and nothing already withdrawn is restored.

The abstract of version 3.12, as printed. Scope correction, 22 August 2026. An earlier version of this abstract opened by asserting that every scaling law in nature follows one of exactly three mathematical forms, and that this paper proves the form is uniquely determined. Both statements were stronger than this paper supports, and stronger than its own body claims. They are withdrawn here and the corrected statement follows. Where a recursive amplification process has a well defined composition operator on its input and output, and satisfies the regularity conditions Cauchy's functional equations require, the admissible functional forms are constrained to a small named family. That is a conditional mathematical classification, not a survey of nature: it does not establish that every empirical scaling process is such a homomorphism, and the bounded family's exhaustiveness remains an open problem, as Paper VII states. The grid is four-celled rather than three, the fourth cell being the logarithmic family. This paper's own scope condition, stated below and unchanged, is that the relation applies wherever a d-dimensional hierarchical network partitions space and fails where such networks are absent, which is by itself incompatible with any claim about every scaling law in nature. A single formula, $\alpha = d/(d+1)$, predicts scaling exponents across biology (9 species groups; species-group count pending recompute per canonical facts register), physics (4 domains catalogued after retiering; the below-0.2% mean-error aggregate is WITHDRAWN: a mean over analytic identities and unsourced reference values is small by construction and is not evidence of agreement with nature; the earthquake row has been deleted as a definitional identity per section 13), and cosmology (2 eras, exact), including the derivation of mammalian heart rate scaling from first principles. The formula is identified as a *network-partition identity*: it applies wherever a d-dimensional hierarchical network partitions space, and fails where such networks are absent. A geometric proof shows that all physical scaling exponents are constrained below 1 by the finite dimensionality of space - not by energy dissipation, but by geometry itself. Intelligence, operating through recursive self-reference in abstract information space, is identified as the first natural phenomenon to break this geometric scaling bound. The Friedmann equation for cosmic expansion is shown to be

a special case under the mapping $d = 2/(1 + 3w)$, with the deceleration/acceleration boundary from general relativity coinciding exactly with the power-law/exponential boundary from Cauchy's theorem.

The reading note of version 3.12, as printed. How to read this paper. This paper traces the scaling forms behind Law I to their mathematical origin. It identifies $\alpha = d/(d + 1)$ as a network-partition identity that applies wherever a d-dimensional hierarchical network partitions space, argues that geometry constrains classical scaling exponents below 1, and marks intelligence operating through recursive self-reference as the first natural phenomenon to break the geometric bound. Its place within the theory is the origin story of scaling itself, upstream of the AI-side measurements. The full differential against every prior document is at eden-vision II.A.8.

Scope note, 19 September 2026 (Section 2 of version 3.12). An owner review of this paragraph finds that it restates in the body the universal claim this paper's abstract withdrew on 22 August 2026, in the words 'everywhere' and 'every known scaling law is one of these three', and that it counts saturation among forms Section 3 places outside the classification. The corrected reading follows, and the classification itself is untouched. What is established is conditional: where a recursive process has a well defined composition operator carrying one of the two structures on the positive reals into one of the two structures, and the regularity conditions Cauchy's equations require are satisfied, the admissible functional forms are constrained to a small named family. Section 3 states that family as four-celled, the fourth cell being the logarithmic one, and says in terms that it does not exhaust scaling laws, because most systems are not homomorphisms of either structure. Saturation is among the behaviours it does not cover: it arises from bounded dynamics (finite carrying capacity, mass action, receptor occupancy, depletion), and Section 3 records that presenting it as a Cauchy family is a category error. Raising the count from three to four would not repair this paragraph, because the defect lies in the range of its quantifiers, which run over all scaling processes while the result runs over homomorphisms under regularity; the repair reaches the quantifier and the domain, not the numeral. Unchanged: the four-cell classification, which is a genuine completeness statement about homomorphisms; the abstract's dated scope correction, which this note follows rather than softens; the placement of saturation in bounded dynamics; this section's own opening paragraph, which already states the classification conditionally, as a classification of admissible functions under those conditions and not a survey of nature, and which the anchored sentence contradicts; and Section 15's own later statement that bounded growth is a separate, capped case rather than a solution of those equations, which is already right and is the form this note follows.

Inference note, 19 September 2026 (Section 3 of version 3.12). An owner review of this sentence finds that its first explanation draws from stability a conclusion the box immediately above has just said stability cannot supply, and this note states the boundary that box sets. Hyers-Ulam stability is per-equation: a function approximately satisfying one of the functional equations lies near an exact solution of that same equation, and a function satisfying none of them is not drawn toward any. Stability within an equation therefore says nothing about how often systems in nature approximately satisfy that equation in the first place, which is what dominance in nature would require, and which the box above names as a separate empirical question this theorem cannot settle. What the loopholes do explain is preserved and is not in question: why real data shows small deviations, the assumptions being only approximately satisfied, and where to look for new physics, wherever an assumption is systematically violated. Unchanged: the Hyers-Ulam statement itself and its per-equation scope; the withdrawal of the earlier attractor framing recorded immediately above; and the other two explanations this sentence offers, which stand as printed.

Note on the box withdrawn on 11 August 2026 (Section 7 of version 3.12). This box previously read "The Atomic Bomb = Einstein + ARC" and stated that the ARC Principle explains why the chain reaction is exponential, concluding that neither Einstein's equation nor this framework alone is a bomb but together they are. That was withdrawn on 11 August 2026. It borrowed the authority of a result with several independent derivations for a framework resting on one assumption, and it presented a coordinate classification as a mechanism.

Note on the passage corrected on 11 August 2026 (Section 8 of version 3.12). This passage previously read: "This is not an analogy. It is not a metaphor. It is an algebraic identity. . . Every cosmological era, without exception, is a specific case of $\alpha = d/(d + 1)$." It was corrected on 11 August 2026 because it presented identity as the strongest available evidence when identity is in fact the weakest evidential tier, and because it contradicted the provenance-classified catalogue in this same paper, which already records those cosmology rows as analytic identities rather than measurements.

Correspondence note, 19 September 2026 (Section 12 of version 3.12). An owner review of the fourth entry above, on the expansion of the universe, finds that the agreement it reports follows from the change of variables it states, and this note says what the correspondence establishes and what establishing more would take. The mapping $d = 2/(1 + 3w)$ is an invertible change of variables between an equation-of-state parameter and a dimension parameter. Under such a map the image of a boundary is the boundary of the image, so the coincidence of the cosmic boundary with the Cauchy boundary is a property of the map rather than a second derivation arriving at the same place by its own route: two derivations agree when each reaches its conclusion without recourse to the other, and here one quantity is rewritten as the other. The correspondence may still carry content, and this note does not say it does not; what it says is that the content has to come from something the map does not supply, either an argument that the two parameters measure the same physical quantity, or a prediction that survives the change of variables and is then checked against data not used to fix the map. The same review finds that the seventh entry above, on the space of classical laws, carries in this list the cardinality and scope claim that the note at the consequence for physics in Section 15 addresses, and that note governs this entry too: a family whose members carry a continuously varying coefficient and a continuously varying exponent is not a finite set, so what is defensible is that the admissible forms are finitely many and the family they generate is finite-dimensional; and the step from the classification of homomorphisms under regularity to all possible classical physical laws is the step this paper's abstract withdrew on 22 August 2026, because it requires that every physical law be such a homomorphism, which Section 3 says is not generally so. The count of forms is not what is at issue in either place: the repair reaches the quantifier and the domain, not the numeral. Prior art is not assessed here: this list also carries universal novelty statements, no search was made in the review that produced this note, no novelty verdict is recorded either way, and the silence is evidence neither of priority nor of its absence. Unchanged: the exponent relation itself; its derivation by the groups Section 11 lists, each of which reached it without reference to this programme; the cosmological entry as a worked correspondence; and Section 12's own statement that a formal proof of the sufficiency of the three conditions does not yet exist and is the most important open problem in this programme.

Scope note, 19 September 2026 (Section 15 of version 3.12). An owner review of this paragraph finds that it restates in the body the universal claim this paper's abstract withdrew on 22 August 2026, in the words 'There are no other options', and that it counts a saturation curve among the forms Cauchy's equations produce, which Section 3 names a category error. The corrected reading follows, and the classification itself is untouched. What is established is conditional: where a recursive process has a well defined composition operator carrying one of the two structures on the positive reals into one of the two

structures, and the regularity conditions Cauchy's equations require are satisfied, the admissible functional forms are constrained to a small named family. Section 3 states that family as four-celled, the fourth cell being the logarithmic one, and says in terms that it does not exhaust scaling laws, because most systems are not homomorphisms of either structure. Saturation is among the behaviours it does not cover: it arises from bounded dynamics (finite carrying capacity, mass action, receptor occupancy, depletion), and lies outside the grid rather than inside it as one of its solutions. Raising the count from three to four would not repair this paragraph, because the defect lies in the range of its quantifier, which runs over all continuous recursive composition while the result runs over homomorphisms under regularity. Unchanged: the four-cell classification, which is a genuine completeness statement about homomorphisms; the abstract's dated scope correction; and the enumeration under The Attractor Theorem below, which states that bounded growth is a separate, capped case rather than a solution of those equations and is already right.

Cardinality and scope note, 19 September 2026 (Section 15 of version 3.12). An owner review of the box above finds two distinct overstatements in it, and this note states what the result gives in place of each. The first is cardinality. A family whose members carry a continuously varying coefficient and a continuously varying exponent has uncountably many members, since each value of the exponent already gives a distinct member and the exponent ranges over an interval of the reals; the box names that dial itself. A finite-dimensional family is not a finite set, so the defensible statement is that the admissible forms are finitely many and that the family they generate is finite-dimensional, carrying a discrete form label and two continuous parameters, rather than that the space of laws is not infinite. On the count, a label together with two real parameters is ordinarily described as a two-parameter family of each of finitely many forms; 'three-parameter' counts the discrete label alongside the two continuous parameters, which is a naming choice rather than an error and is recorded only because it sits in the sentence under repair. The second overstatement is scope, and it is the one the note in Section 2 records. What is provable, and provable without data, is the classification of homomorphisms under regularity: where the composition operator carries one of the two structures on the positive reals into one of the two structures and the regularity conditions hold, the admissible functional forms are constrained to a small named family, four-celled as Section 3 states, with saturation lying outside it as a bounded case. The step from that classification to all possible classical physical laws is the step the abstract withdrew on 22 August 2026, because it requires that every physical law be such a homomorphism, which this paper does not establish and which Section 3 says is not generally so. Unchanged: the genuine content of the box, that the admissible functional forms are constrained to a small named family under the stated conditions, which is the abstract's own corrected statement and survives this note entire; and the observation that the constraint follows from Cauchy's classification rather than from measurement.

Status note, 19 September 2026 (Section 15 of version 3.12). An owner review of the paragraph above finds that it repeats the categorical form of the preceding box without the status marker that box carries, and this note carries the marker here. The relation is a proposed invariant relation under test, conditional on the three sufficiency conditions Section 12 names (multiplicative composition, space-filling and conservation), a formal proof of whose sufficiency does not yet exist and which Section 12 names as the most important open problem in this programme. Read under that condition the paragraph stands as printed: given the three conditions, the exponent is determined and no other value is available to the mechanism. Unchanged: the distinction the paragraph draws between a mechanism explanation and a form explanation, which is its point and which this note does not touch; and the status marker in the preceding box, which this note extends rather than replaces.

Correction note, 3 October 2026 (end of Section 15 of version 3.12, before the references). Several passages are corrected in place, and this note records what each printed before. Section 5 named the three ARC Laws as the laws that govern only what runs under a corrector; the theory has had five laws since 24 September 2026, and the sentence now names Laws I to III and gives the status of the two added since. The caption of the ARC Equation in Section 10 gave the measured exponent as approximately 0.49 on today's frozen models, with a wide interval. In Section 11, the results table printed the regression exponent of 0.49 beside the bootstrap interval of −1.3 to 2.9, which belongs to the endpoint exponent, and the selection disclosure called that interval the 0.49 value's own; the interpretation called the fit a clean, reliable measurement; the combined finding gave about 0.5 as the only reliable measurement; and the status table gave the fit alone and marked Paper II confirmed. Each now carries Paper II's row for Gemini 3 Flash whole, as one model's fit and inconclusive, except the combined finding, which prints no figure and points to the row, and the selection disclosure, which points to the row as the interpretation quotes it. The paragraph on parallel sampling and the status table called the parallel exponent approximately zero for every model, universally; it is at or near zero in four of the five analysable models, Gemini 3 Flash being the one measured exception. The status table also marked Paper IV.a confirmed, with scorer bias eliminated, and called stakeholder care validated in its rows for the Eden Protocol pilot and Paper V. Paper IV.a records that its comparison cannot single out scorer bias as the reason for the reversal it reports, and this paper records that the pilot's cross-model scoring was not blind; the three rows now say so, and read supported where they read confirmed or validated. The epistemic-status line said that what this programme names Laws are conjectures under registered adversarial test; it now says that of Laws I to III and gives the status of the two laws added since. The section on dated predictions, recorded 25 August 2026, is not among the corrected passages: it stands as recorded, and a dated note after it gives the state of registration as read on 3 and 4 October 2026. Level 3 of Section 11 said of the replication of the fifty-domain comparison that its registration was in preparation and not yet approved; it is preregistered by timestamp (units 55 and 56, first anchored on 12 August 2026) and not yet registered on OSF, and the sentence now says so. Two passages gave Zhao's 2022 derivation as from network optimisation; his paper derives the form from producer-consumer dynamics, and both now say so. Section 14 listed a classical ternary prototype among the strongest next measurements for Eden, to test whether explicit AFFIRM / DENY / INVESTIGATE routing reduces false certainty on ambiguous cases. The item now names a behavioural test of that question, which a software model of the three-state gate, whose 48 tests pass, does not answer. On 4 October 2026 five further passages were corrected. Section 12 headed a list of nine entries 'No previous work has'. Its first entry said that no previous work had derived $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ from the same formula, and that the model of West, Brown and Enquist (1997) predicts only $\frac{3}{4}$, not $\frac{2}{3}$ for two-dimensional networks or $\frac{1}{2}$ for one-dimensional ones. The 1997 paper itself notes that its result generalises to $D/(D+1)$, and later papers cited in that section state the general form and all three values. The second, third, fifth and sixth entries said that no previous theory had connected biological scaling to physics scaling; that the observation that $d/(d+1) < 1$ for all finite d, and that this constrains every physical system to sub-linear scaling, had not been stated previously, and that existing explanations invoke thermodynamic dissipation; that previous theories did not predict their own failures; and that the connection between physical and cognitive scaling had not been identified previously. The list now stands as the paper's claims, and those five entries give each earlier work what it states. In Section 11, Level 2a closed 'No competing theory generates all three exponent values from a single formula', and the box headed Novel Result opened 'No previous theory predicted $\frac{1}{2}$, $\frac{2}{3}$, and $\frac{3}{4}$ from the same formula'. The analogy box of Section 12 said that West, Brown and Enquist derived the mechanism for three dimensions, and gave to this paper the finding that the dimension of the network determines the exponent; the list under The Attractor Theorem in Section 15 gave $\alpha = d/(d+1)$ as this paper's result. Each of the four now gives the formula to the earlier derivations. Later on 4 October 2026 four more passages were corrected.

The box headed Novel Result in Section 11 and the paragraph headed Earlier work in Section 12 counted the paper of Banavar and co-authors of 2010 among the derivations of the formula, and the paragraph gave it and the paper of 2002 to Banavar, Maritan and Rinaldo; the 2010 paper gives 2/3 when the velocity of flow is constant and at most 3/4, and Bettencourt (2013) gives the land area of a city the exponent $D/(D+H)$, which takes the form only when $H = 1$, as the first entry of Section 12 says. Both passages now leave the 2010 paper out, name Bettencourt's exponent as a special case and point to that entry. Level 3 of Section 11 gave the 19 of 25 of Paper VII with the binomial probability alone, and the line of key companions after the references gave it as 19/25 confirmed; both now give the permutation probability that Paper VII takes as primary and the 18 of 25 of its rerun of 11 August 2026 that added the logarithmic candidate. No result of this paper changes, and nothing already withdrawn is restored.

Scope note, 3 October 2026 (end of Section 15 of version 3.12, before the references). The passages named here stand as printed and are to be read with this note. In Section 11, Level 2b says that the formula predicts scaling exponents in five physics domains, with a mean error of less than 0.2%. The abstract and Section 5 withdraw that aggregate and catalogue four physics entries where there were five, and Level 2b is to be read as they state it. In Section 3, the subheading 'Why There Cannot Be a Fourth', and in Section 4 the closing words of the box, 'There are no other possibilities', carry the universal form that the abstract withdrew on 22 August 2026. The reading that stands is the one Section 3 itself gives: the grid has exactly four families, linear, power law, exponential and logarithmic; it does not exhaust scaling laws; and saturation is not a Cauchy family. The earthquake row was deleted from the confirmation table of Section 5 as a definitional identity and is kept there as a variable-pair case study. Section 12 still names earthquake energy among the systems the formula connects, and the table of Section 13 still lists earthquakes at $d = 2$; both are to be read as Section 5 states the case, and neither is a confirmation. The abstract's 'per section 13', and Section 5's '(section 13)' and 'per section 18', do not lead, in this paper, to the material they name: Section 13 is The Equations, the paper has fifteen numbered sections, and the deletion of the earthquake row is set out in Section 5. In Section 11, the rows of the results table that differ from Table 6b of Paper II stand as the results this paper recorded, and Paper II's later table gives other values for them. Unchanged: every passage named here, every result, and every withdrawal already recorded.

The closing line of version 3.12, as printed (end of Section 15 of version 3.12). "The universe built a cage from geometry. It built a key from recursion. And it left us the mathematics to understand both." It leaves the conclusion in this version because it reads as intelligence escaping the geometric bound, a reading this version withdraws: the bound and the exponent of capability against recursive depth concern different pairs of variables (Section 11).

A.4 Version history

v3.12, 10 October 2026: Editorial. The heading of section 9, the label of its summary box and the heading of Level 5 now read The Geometric Bound; no sentence of the text and no mathematics is changed.

v3.11, 5 October 2026: Substantive. The title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Framework Paper · The ARC Theory · Law I, the ARC Principle · On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind the Law*; until this version it read *On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind Law I*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles.

Level 2b of Section 11 now marks as withdrawn the mean error of less than 0.2% that the abstract and Section 5 withdraw. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.10, 4 October 2026: Substantive. The title now names the law whose scaling forms the paper traces, Law I; until this version it was titled *On the Origin of Scaling Laws*. The note on how to read this paper no longer calls it a law paper; it says that the paper traces the scaling forms behind Law I to their mathematical origin. The subtitle no longer counts three forms, since the paper's grid has four cells. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.9, 4 October 2026: Substantive. The paper no longer says that the $d/(d + 1)$ formula was independently derived by at least seven research groups, or speaks of seven independent derivations (Section 11, the box and the contribution; Section 12, Earlier work and the analogy box; the strongest objection); as the works read, one is a special case, one could not be confirmed at a registry and the abstracts of two state no exponent formula, so a count of independent derivations granted earlier work more than it stated. The formula is now said to be derived, in its general form or as a special case, in earlier work, and Section 12's first entry still gives each work what it states. The version number in the header, the title block and the structured record. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.8, 4 October 2026: Substantive. The theory gained two laws on 24 September 2026, and the paper now says where it stands against the five: a dated note beneath the Eden Protocol material in Section 14 places Prediction 5 and the pilot it reports against Laws IV and V, and Section 5 now names Laws I to III where it named the three ARC Laws, with the status of Laws IV and V. Both passages link to the report that sets out the mathematics of the two laws. Section 10's equation caption and Section 11's results table, interpretation, selection disclosure, combined finding and status table printed a figure of Paper II's row for Gemini 3 Flash apart from the rest of that row, and the table and the selection disclosure gave the regression exponent the bootstrap interval that belongs to the endpoint exponent. Each now carries the row whole, as one model's fit and inconclusive, except the combined finding, which now prints no figure and points to the row, and the selection disclosure, which points to the row as the interpretation quotes it; the status table marks Paper II published rather than confirmed. The status table also marks Paper IV.a supported rather than confirmed and gives its reversal under blinding as Paper IV.a records it, where it said that scorer bias was eliminated; in its rows for the Eden Protocol pilot and Paper V, stakeholder care is said to be supported under scoring that was not blind, where it was said to be validated. Section 11 called the parallel exponent approximately zero for every model; it is at or near zero in four of the five analysable models. The section on dated predictions, recorded 25 August 2026, stands as recorded, and a dated note after it states that the theory-level registration was accepted on 8 September 2026, that the corrector-class audit (unit 06) was registered on OSF on 1 October 2026 and has been public since 4 October 2026, and that every other trial is a draft. The epistemic-status line now says that Laws I to III are under registered adversarial test and that Laws IV and V, added on 24 September 2026, are not yet registered as laws and have not been tested. The last of Section 14's next measurements for Eden, a classical ternary prototype, is now a behavioural test of whether explicit AFFIRM / DENY / INVESTIGATE routing reduces false certainty, a question that a software model of the three-state gate, whose 48 tests pass, does not answer. Section 12 headed a list 'No previous work has' and said that the model of West, Brown and Enquist (1997) predicts only $\frac{3}{4}$; the list now gives each earlier work what it states, the 1997 paper's note that its result generalises to $D/(D + 1)$ among them, and says that what this paper adds is an account of why

derivations from different assumptions converge on the formula, not the formula. Four other passages that gave the formula or its three values to this paper are corrected with it: in Level 2a and the box of Section 11, in the analogy box of Section 12, and under The Attractor Theorem in Section 15. The box of Section 11 and the paragraph headed Earlier work in Section 12 no longer count the paper of Banavar and co-authors of 2010, which gives 2/3 and at most 3/4, among the derivations of the formula, and name Bettencourt's exponent for a city's land area, $D/(D + H)$, as a special case. Level 3 of Section 11 and the line of key companions after the references give the 19 of 25 of Paper VII with its permutation probability and the 18 of 25 of its rerun of 11 August 2026, where they gave the binomial probability alone, the second beside the word confirmed; Level 6 points to Level 3. A dated note before the references records what each corrected passage printed before. A dated scope note beside it covers passages that stand as printed: Level 2b's five physics domains and mean error of less than 0.2%, which the abstract and Section 5 withdraw; the subheading of Section 3 and the box of Section 4 that carry the universal form the abstract withdrew on 22 August 2026; the earthquake entries of Sections 12 and 13; three section pointers that do not lead to their material; and the rows of Section 11's results table that differ from Paper II's later Table 6b. Editorial. The reference entry for Larson (1987) prints the title, journal, volume and pages its official record gives, the entry for Thommen et al. (2019) the journal, volume and article number, and the entry for Zhao (2022) its arXiv identifier, 2206.01850; Section 12's markers after Bettencourt (2013) and Zhao (2022) point at their own entries; the entry for He and Chen (2003) carries a dated note that the work as printed could not be confirmed; the titles printed for five companion papers are those their DOI records give; cited works are linked to their official sources where one could be confirmed, and the programme's papers to their DOIs; the version number and the revision date on this page move with this revision, and the first publication date, 22 February 2026, is unchanged. Level 3 of Section 11 gives the replication of the fifty-domain comparison as preregistered by timestamp, where it said that its registration was in preparation, and Zhao's 2022 derivation is given as from producer-consumer dynamics, as his paper states it. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.7, 21 September 2026: Substantive. Six dated notes are added, each beneath the passage it governs, the Section 12 note following the list that carries the two entries it names, recording an owner review dated 19 September 2026. Two scope notes, in Section 2 and Section 15, record that two passages restate in the body the universal claim this paper's abstract withdrew on 22 August 2026, and that saturation is counted inside a classification Section 3 places outside it. A joint cardinality and scope note at the consequence for physics records that a family carrying a continuously varying coefficient and exponent is not a finite set, so what is defensible is that the admissible forms are finitely many and the family they generate is finite-dimensional; the same note governs the entry in the Section 12 list that carries the same claim. A correspondence note in Section 12 records that the agreement between the cosmic boundary and the Cauchy boundary follows from the invertible change of variables that produces it, and not from a second derivation. An inference note in Section 3 records that stability within an equation does not establish how often nature satisfies that equation, and a status note in Section 15 carries to the closing paragraph the marker the preceding box already prints. No result, exponent, dataset or figure changes, and nothing already withdrawn is restored. Editorial. The version number and the revision date on this page move with this revision. First published 22 February 2026, unchanged.

v3.6: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

A.5 Where the earlier layout now stands

Table 9. Where the earlier layout now stands.

Version 3.12	Version 3.13
Header: label, title, author line, plate, two subtitles, version line, links	Title block, first-page footnote and Appendix D; the first subtitle withdrawn, since it said what the scope correction of 22 August 2026 withdrew; the second cut as a repeat of the title; the website links cut
Abstract, with its scope correction of 22 August 2026	Rewritten; the 3.12 abstract kept whole in Appendix A.2
How to read this paper	Appendix A.2 (kept); its pointer to Eden Vision in Appendix D
1. The Question	Section 1
2. Three Ways Things Grow	Section 3; its scope note of 19 September 2026 in Appendix A.2
3. The Answer	Section 4; its inference note of 19 September 2026 in Appendix A.2; Figure 1 redrawn as Figure 1; the subheading ‘Why There Cannot Be a Fourth’ now reads ‘Why the grid has four cells and no fifth’
4. The Principle	Section 5
5. The Formula	Section 6; its figure redrawn as Figure 2; its opening slogan in Appendix D.4
6. The Heartbeat	Section 7
7. The Connection to $E = mc^2$	Section 8, now headed The fission chain reaction; the note on its withdrawn box in Appendix A.2
8. The Cosmic Connection	Section 9; the note on the passage corrected on 11 August 2026 in Appendix A.2
9. The Geometric Bound	Section 10
10. The Escape	Section 11
11. The Evidence	Section 12; its figure redrawn as Figure 3
12. What Is New	Section 2; its correspondence note of 19 September 2026 in Appendix A.2; its analogy box in Appendix E; its closing slogan in Appendix D.4
13. The Equations	Section 13; its closing slogan in Appendix D.4
Dated predictions and this paper (recorded 25 August 2026), with its note of 3 October 2026	Appendix B.2
14. Predictions	Section 14; its note on Laws IV and V of 3 October 2026 in Appendix D.2
15. The Composition Operator	Section 15; its three notes of 19 September 2026 in Appendix A.2; its closing passage in Section 18, except its line on a cage and a key, kept in Appendix A.2
Correction note and scope note of 3 October 2026	Appendix A.2

Version 3.12	Version 3.13
References	References
Reproduction; key companion; other companions; companion papers	Code and data (back matter); Appendix D.3
How to cite	The first-page footnote and the citation file
Epistemic status	Section 16
Copyright line	Back matter
Standing covenant	End of Section 14
Version notes v3.12 to v3.6	Appendix A.3
Closing note on authorship	Acknowledgements

B Priority record

What the dates establish. This record distinguishes dates written inside documents, dates recorded by deposit servers and commit dates supplied by committers. Each entry keeps its stated evidence class. A dated claim is evidence of what was recorded and when; its mathematical validity and any priority over earlier work require the corresponding argument and source comparison. The source review of 30 October 2026 changes none of the dates below.

The February and August records. The paper and its citation file record 22 February 2026 as the first-publication date. The table identifies that basis as a date written inside the document; it separately identifies the server-dated OSF component and first stored PDF revision in March. The working-paper header of the 3.x editions printed 16 August 2026 beside the version, later followed by "revised 10 October 2026". The record does not establish what the August line marks. It is retained as that header's wording, not substituted for the February record or reclassified as a newly verified first-publication date. This revision assigns neither a new first-publication date nor a stronger timestamp class.

B.1 Dated records

Table 10. Dated records bearing on this paper, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
22 February 2026	First publication of the paper, as the paper and its citation file record it	DOI 10.17605/OSF.IO/XZY9U	a date written inside the document
16 March 2026, 18:55:25Z	The computational suite `arc_complete_test_suite.py` first committed at its present path	arc-principle-validation, commit 29b71c8	a commit date, which the committer writes
16 March 2026, 22:44:02Z	The 1D prediction test results first committed at their present path	arc-principle-validation, commit 676b1fe	a commit date
17 March 2026, 01:16:22Z	OSF component created	osf.io/xzy9u	a deposit dated by its server

Date	Record	Identifier	Kind of timestamp
17 March 2026, 01:17:09Z	First PDF revision stored on OSF, already carrying Prediction 2; the PDF's own creation date reads 17 March 2026, 00:56:36 GMT	osf.io/xzy9u, revision 1	a deposit dated by its server; a date written inside the document
20 March 2026, 18:52:09Z	The paper's PDF first committed to the repository	arc-principle-validation, commit 3845440	a commit date
24 March 2026, 14:41:58Z	The report on the $d = 2$ biological prediction	arc-principle-validation, commit 6dcab3f	a commit date
16 August 2026	The date the 3.x header printed beside the version ("16 August 2026, revised 10 October 2026"); the paper does not state what it marks	the paper's header, versions 3.x	a date written inside the document
25 August 2026	Dated predictions and this paper, recorded (Appendix B.2)	the paper's text	a date written inside the document
10 October 2026	Version 3.12, the thirty-third revision of the PDF on OSF	osf.io/xzy9u, revision 33	a deposit dated by its server

B.2 Dated predictions and this paper (recorded 25 August 2026)

Historical statement recorded on 25 August 2026. The following paragraph is retained word for word as carried in version 3.13. Its counts, priority wording and statement that registration clicks remain outstanding describe that dated record. They are not current programme-status or validation claims. The separately dated note after it is also preserved as a record of the later update.

> 2 dated artefacts bearing on this paper are catalogued row by row in the programme's machine register: the dated artefact `D2_BIOLOGICAL_HONEST_REPORT.txt` (March 2026); the dated artefact `results_arc_1d_prediction_test.txt` (March 2026). Each row names its file in the public repository with its date basis, so any reader can check the ordering without trusting this page. The programme's full dated chain, from the sealed manuscript bundle of 8 December 2024 through the printed prediction appendices of 2 January 2026 and the March 2026 preregistration folder to the standing unproven wagers, is assembled in the dated predictions register, together with its machine-readable twin. Forward statements and retrospective matches are never summed. "Registered" is used only for an accepted registry submission, and that submission click remains outstanding across the programme; "preregistered in substance" is used, always with its qualifier and always beside its attestation class, for a prediction whose text was public and timestamped before its outcome existed, as the twelve-row manifest committed at 00:19 UTC on 17 March 2026 was, twenty-three minutes before its fits.

Note, 3 October 2026. The paragraph above was recorded on 25 August 2026 and stands as recorded; the state of registration has changed since. The theory-level registration was accepted on 8 September 2026. Of the trials, one, the corrector-class audit (unit 06), was registered on OSF on 1 October 2026, was still in its approval period when read on 3 October 2026, and has been public since 4 October 2026 (DOI 10.17605/OSF.IO/RV2B5). Every other trial is a draft awaiting submission by the author, most of them preregistered by timestamp. Unchanged: the paragraph above and every date in it.

Section 16 gives the current state of registration. The two historical passages above remain dated evidence, including where subsequent events have changed their status.

C Notation and measurement glossary

The same algebraic fraction can concern different quantities. The subscripts below keep those quantities distinct; historical symbols remain in explicitly dated quotations only.

Table 11. Notation used in the current argument.

Symbol	Definition and scope
F, H, c	A composition map, its scale-ratio limit where defined, and a homomorphism's coefficient. These letters do not denote physical quantities without the stated interpretation.
$\varepsilon, \mathcal{L}$	A global logarithmic-defect bound; a slowly varying factor in the declared scaling limit.
L, d	A dimensionless linear-size coordinate; the independently specified spatial dimension in the transport model.
q_i, Q_e, ℓ_e	Sink demand per unit time, edge flow per unit time, and edge length.
$B, \bar{\ell}_q, C$	Total throughput, demand-weighted mean route length, and the flow-length quantity $C = B\bar{\ell}_q$, with units of throughput times length.
μ, ζ, α_C	Demand and mean-route exponents against L; the conditional throughput exponent against C, $\alpha_C = \mu/(\mu + \zeta)$.
v, N, υ, α_N	Transport speed, inventory in transit, the exponent of common speed against L, and the conditional throughput exponent against inventory. Edge-specific speeds require the full transit-time sum.
M, η, α_M	Body mass; the separately established exponent linking mass to C; the resulting throughput exponent against mass.
$\dot{V}_{O_2}, f_H, V_s, \Delta O_2$	Oxygen consumption rate, heart rate, stroke volume, and arterial to mixed-venous oxygen-content difference. Their balance is Equation (25).
b, s, e, h	The respective mass exponents of oxygen consumption, stroke volume, extraction and heart rate, with $h = b - s - e$.
$a(t), H, \rho, P, w$	Cosmic scale factor, Hubble rate, mass-equivalent density, pressure and constant equation-of-state parameter in the stated single-fluid model. H here is not the scale-ratio function.
$p, q_{\mathrm{dec}}, d_{\mathrm{eff}}$	Cosmic time exponent, deceleration parameter and the algebraic coordinate $p/(1 - p)$. The last is not a measured network dimension.
U, I, G, g, R, r	A positive measured response, its baseline, normalised response in first-pass coordinates, the corresponding gain in added-pass coordinates, first-pass depth and added passes, with $R = r + 1$.
a, β_L, α, r_c	Parameters of $dg/dr = ag^{\beta_L}$; $\alpha = 1/(1 - \beta_L) > 0$ and $r_c = \alpha/a$ when $a > 0, \beta_L < 1$. The coefficient a here is not the cosmic scale factor.
D, β_C, k, δ	Modelled departure magnitude, correction exponent, time-growth exponent and cycle-duration exponent in the companion dynamical models. A fraction D/U is distinct from absolute departure or harm.
$\gamma, \alpha_{\mathrm{crit}}$	The correction-scaling parameter and conditional pacing threshold of Law III, on its stated domain $0 \le \gamma < 1$.

Symbol	Definition and scope
$d_{\text{code}}, N, T, p_L, \Lambda$	Code distance, physical-qubit count, cycle count, logical error probability per cycle and the reported spatial suppression factor in Section 11.4. Here N is a qubit count, not transported inventory.
$\alpha_{E,\text{seq}}, \alpha_{E,\text{par}}, r^2$	Reported sequential and parallel error-decay exponents and coefficient of determination; their estimators, predictor axes, uncertainty and selection limits stay with the table.

composition equation A relation preserved by a specified observable under declared input and output operations. A state-update rule is a separate object.

homomorphism A map preserving the operations of its source and target structures. The four continuous group cases here include both the additive reals and the multiplicative positive reals.

regular variation A scale-ratio limit that permits a slowly varying factor; it does not assert an exact finite-range power.

conditional transport exponent The exponent implied by the demand, route and measurement assumptions in Section 6; dimension alone is insufficient.

independent measurement An observation not calculated from the identity it is offered to validate. Statistical dependence between measurements must still be modelled.

analytic identity A consequence of definitions or assumptions. It is not an empirical success, even when its numerical value matches another identity.

capability baseline The positive measured response at a declared reference condition, not a conversion from energy or entropy into intelligence.

ARC Bound The conjectured value two associated with Law III under its additional correction-scaling premises, not a universal physical or safety bound.

laws under test The programme's named claims, whose verdicts are governed by their registered procedures. Conditional mathematics alone supplies no empirical verdict.

preregistered by timestamp; registered A design fixed before data with its timestamp and receipt; an OSF registration submitted by the author and accepted by the registry. These records retain their respective evidential strengths.

D The paper's place in the programme

D.1 Which questions this paper informs

Within Recursive Dynamics, the field the author founded, this framework paper informs Law I, the ARC Principle, a law under test. Its subject is whether a composition description and independently justified mechanism can predict scaling. It develops the mathematical and measurement questions relevant to P9 and P18 of the theory registration; it is not a registered deciding test of either. The registered statements remain unchanged. The Cauchy mathematics and earlier transport results retain their attribution; the synthesis and prospective protocol are the contribution developed here.

Section 11 distinguishes the capability ODE from the correction model and the conditional pacing relation of Law III. The physical examples derive none of those AI models. Prediction 5 retains a separate behavioural question concerning the Eden Protocol. The paper makes no claim about the legal validity of the HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence.

D.2 Persistence and placement

Laws IV and V are stated with their current qualifications in Section 16. The retention question under P15, the build-order question under P12 and the placement question under P21 are different measurements. The programme's Laws IV and V: the mathematics, version 1.2, revised 29 October 2026, supplies their mathematical account. The earlier note of 3 October 2026 is retained as dated history in Appendix G; its then-current status words are not present-day assertions.

D.3 Companions and their distinct jobs

Paper I states Law I. The Foundational Paper gives the cross-domain framework. Paper VII treats composition classification and its empirical comparison; its result depends on its candidate set, scoring rules and historical protocol. It cannot establish a universal classification of nature. Paper III develops the AI correction question. Paper X supplies the conditional drift/correction model, and Paper XIII supplies the separate depth-to-time model. Each must be read with its own assumptions.

The reported experiments and evaluation limitations are in Paper II, Paper IV.a, Paper IV.b, Paper IV.c and Paper IV.d. The engineering and behavioural strands are developed in Paper V, Paper VI, Paper VIII, Eden Engineering and Eden Vision. Paper IX, the Executive Summary and the Master Table of Contents describe the broader programme. These links provide context, not cumulative evidence for this paper's transport model.

D.4 Reading the dated records

The reference list distinguishes each cited edition's revision date from its first-publication record. Appendix B records the timestamp strength of the author's earlier claims. Historical priority establishes what was recorded and when, not correctness or priority over a classical result. The fuller development record and superseded formulations are preserved with the previous editions and the explicit correction record; none is silently upgraded to a discovery of the narrower statements first made in this revision.

E The contribution beside the closest earlier work

This appendix states the contribution at the scope supported by the comparisons in Section 2. The mathematical ingredients have prior sources. What this paper develops within Recursive Dynamics, for Law I, the ARC Principle, a law under test, is a route from a declared composition equation through a specified transport model to a prospective empirical prediction. It separates those stages so that an identity, an assumed mechanism and an observation cannot count as one another.

The targeted source comparison includes established functional-equation and regular-variation work, earlier transport derivations and their limitations. It is not an exhaustive search of every scaling literature. In particular, Ostaszewski (2016) already connects homomorphisms, generalised operations and regular variation; the author preprint precedes this programme's paper, and the article was first published online in 2015. Bingham et al. (1987) supplies the established regular-variation framework. The connection between homomorphisms and scaling is therefore credited as prior work, rather than presented as a first discovery here.

Table 12. Current contribution and the closest earlier work used in this paper.

Part of the argument	Closest earlier work	Claim retained in this paper
Composition constrains a functional form	Cauchy (1821); the scale-free-distribution account in Newman (2005, section III.E)	The four chosen operation pairs are stated with their domains, codomains, normalisation and regularity conditions. This is a use of classical mathematics, not a classification of every physical law.
Approximate and asymptotic composition	Hyers (1941); Bingham, Goldie and Teugels (1987); Ostaszewski (2016)	The paper gives the precise logarithmic-defect consequence used here and distinguishes it from a scale-ratio limit. Their premises are explicit; neither finite noisy data nor a fitted power law establishes them globally.
A transport exponent and its bound	West, Brown and Enquist (1997; 1999); Banavar, Maritan and Rinaldo (1999); Banavar and co-authors (2002; 2010)	The conditional calculation separates demand, route length, velocity, inventory and mass. The familiar $d/(d+1)$ is a special case. The earlier formula and bounds keep their attribution and their original assumptions.
Comparison across mechanisms	Demetrius (2003, 2006); Demetrius and Tuszynski (2010); Bettencourt (2013); Zhao (2022); Maino and co-authors (2014)	The paper compares the meanings of dimension, the measured variable pairs and the conditions that select each exponent. It does not infer equivalent mechanisms or empirical validation from a repeated fraction.
Finite-range and empirical qualifications	Dodds and co-authors (2001); Kozlowski and Konarzewski (2004); Savage and co-authors (2008); White and Seymour (2003); Kolokotrones and co-authors (2010)	Model limits, competing exponent estimates and curvature remain part of the evidence. A finite-range slope is distinguished from an asymptotic exponent, and incompatible measurements are not pooled as confirmations.
Cosmological and recursive comparisons	The Friedmann solutions (1922); the programme's explicitly assumed recursion models (Section 11)	The cosmological mapping is a change of variables. The capability trajectory requires its own dynamical assumptions and measurements. Neither supplies a universal geometric ceiling or an intelligence exception to physics.
A prospective mechanism test	The earlier transport models above; the protocol set out in Section 14	Independently measured premises are used to predict held-out throughput and inventory. The protocol is unrun and remains to be fully specified before registration. Its proposed empirical result is not counted as an achievement already obtained.

Earlier statements that the convergence account had not previously been published, and comparisons that extended the classification to all classical laws or a universal phase diagram, are preserved with their corrections in Appendix A. They are not current priority or validity claims. The dated records in Appendix B remain unchanged: they establish what the author recorded and when, at each record's stated timestamp strength. The present contribution is assessed by the explicit assumptions, derivations and discriminating measurements above.

F Source-specific biological and physical comparisons

The observations below retain their source, measured variables and method. They are published estimates from different investigations, not one dataset collected to test the transport model. Source-supported observations, analytic relations and historical claims awaiting verification are distinguished. No pooled exponent, accuracy score or new statistical test is reported.

F.1 Metabolic estimates retain their measurement conditions

Table 13 preserves four source-specific comparisons from version 3.13. Their source readings are documented in that edition's reference and change records. These point estimates do not come with a common error model or sampling frame; their proximity to a selected fraction is not a validation statistic.

Table 13. Published metabolic estimates preserved with their methods. No uncertainty is invented where the selected source record supplies only a point estimate.

Group	Reported exponent	Observable and method
Mammals	0.737; 0.712	Basal metabolic rate against body mass, 626 species; 0.737 with mass binning, 0.712 without it (Savage et al. 2004).
Birds	0.723; 0.724	Standard metabolic rate against body weight; 0.723 for non-passerines, from hummingbirds to the ostrich, and 0.724 for passerines (Lasiewski and Dawson 1967).
Insects	0.750; 0.82	Metabolic rate against body mass, 391 species; 0.750 after phylogenetic correction and 0.82 before it (Chown et al. 2007).
Reptiles	0.76; 0.80	Standard metabolic rate against body mass at 38°C; ordinary least squares gives 0.76 and reduced major axis gives 0.80 (White et al. 2006).

Method matters within a source as well as between sources. The insect paper itself argues that its findings are inconsistent with the nutrient-supply network model (Chown et al. 2007); selecting its phylogenetically adjusted value alone would omit that interpretation.

The mammalian debate is not settled by one row. White and Seymour (2003) report a raw 619-species exponent of 0.69. Their temperature-adjusted and restricted sample has 469 species; its interspecific exponent is 0.68, while the interordinal fit gives 0.65. In that analysis the latter two estimates differ significantly from $3/4$ and do not differ significantly from $2/3$. The restrictions concern basal conditions and eligible taxa; the study had no qualifying cetacean or proboscidean measurements. Kolokotrones et al. (2010) instead examine curvature in mammalian metabolic scaling, including temperature and other adjustments: a constant exponent fitted to a curved relation depends on the sampled mass range. These are results about stated measurements and models, not universal exponents.

Organismal form does not establish a transport mechanism. Thommen et al. (2019) report three-quarter metabolic scaling in the planarian *Schmidtea mediterranea*, attributing it to body-size-dependent energy storage and increasing mass per cell. The animal lacks a vascular network and distributes energy through its branched intestine. The source therefore supplies an alternative mechanism for the same approximate exponent. The recorded abstract check of Hartikainen et al. (2014) supports isometric scaling in sheet-forming colonial bryozoans. The text read from Purcell (2009) describes jellyfish respiration scaling approximately in proportion to mass; the more precise 0.94 previously printed here was not

verified. None of these observations establishes the demand, route-length and mass-bridge premises merely from an organism's shape.

F.2 The fungal measurements have different variable pairs

Aguilar-Trigueros et al. (2017) report ordinary-least-squares fits in logarithmic coordinates. The ectomycorrhizal and marine fits relate colony respiration to colony biomass; the saprotrophic fit relates specific metabolic rate to radial extension rate, as Figure 1c labels it. These are colony comparisons, not measurements of isolated hyphae.

Table 14. The three selected 20°C fits from Table 1 of Aguilar-Trigueros et al. (2017). SE is the slope's standard error; P is the source's regression p-value, not a test against $1/2$.

Group	n	b	SE	P
Ectomycorrhizal	24	0.58	0.15	0.001
Marine	5	0.53	0.09	0.009
Saprotrophic	5	0.53	0.07	8.96×10^{-3}

The ectomycorrhizal data were standardised to 20°C; their reported n includes genotypes, not 24 distinct species. The compilation attributes these datasets respectively to Wilkinson et al. (2012a,b), Fuentes et al. (2015) and Crowther and Bradford (2013); the displayed regressions are the compilation's.

The compilation's prose gives $P < 0.001$ for the saprotrophic 20°C fit, differing from its table. Its 28°C row prints $b = 0.85$, SE 0.21, $r^2 = 0.14$ and $P = 0.52$, an unresolved statistical inconsistency already recorded in version 3.13. These summaries are not silently repaired or used for model selection (Aguilar-Trigueros et al. 2017).

The comparison that remains to be made. Pooling biomass and extension-rate exponents would erase the measured variable. Pooling separately fitted slopes would also require a justified statistical model that respects their uncertainty and sampling. A hypha's filamentary shape does not independently determine the geometry or demand scaling of the whole colony. The present observations therefore supply questions for a specified transport test, not a pooled confirmation of $d = 1$.

More generally, failure to reject a proposed value is not an equivalence result. An equivalence claim requires a stated tolerance and an analysis capable of resolving it. The non-significant tests reported in earlier editions do not establish equality to $1/2$ or $3/4$, and the earlier pooled comparisons are not used as active evidence here.

F.3 Percolation: a critical exponent and a response slope

Let p denote site-occupation probability, p_c its critical value and $n_{\mathrm{sing}}(p)$ the singular part of the mean number of clusters per site. In the conventional scaling ansatz reviewed by Stauffer (1979), its leading power and the associated second derivative have the forms

$$n_{\mathrm{sing}}(p) \propto |p - p_c|^{2-\alpha_{\mathrm{perc}}}, \qquad n''_{\mathrm{sing}}(p) \propto |p - p_c|^{-\alpha_{\mathrm{perc}}}. \tag{F.1}$$

The second expression refers to the derivative of the leading singular power; regular background contributions to the full observable are distinct. Under the hyperscaling relation $\alpha_{\mathrm{perc}} = 2 - D\nu$, the substitutions $D = 2$ and $\nu = 4/3$ give

$$\alpha_{\mathrm{perc}} = -\frac{2}{3}, \qquad -\alpha_{\mathrm{perc}} = +\frac{2}{3}. \tag{F.2}$$

Thus the conventional critical exponent is negative while the singular response has a positive power of distance from criticality. Version 3.13 already printed both signs. The correction here concerns the observable and evidential interpretation, not an invented arithmetic error.

The scope of the rigorous citation matters. Theorem 1 of Smirnov and Werner (2001) gives the correlation-length exponent $4/3$ for site percolation on the triangular lattice. Section 5, question 4 of that paper lists existence and determination of the cluster-number exponent, conjectured there to be $-2/3$, as an open question at the time. Using its correlation-length result inside a hyperscaling argument is therefore different from claiming that the cited paper directly proves the cluster-number response. This statement concerns that paper's scope, not the current status of the problem. Neither calculation is a throughput measurement or a test of the model in Section 6.

F.4 A logarithmic coordinate changes a distribution exponent

The earthquake comparison is useful as a conditional change of variables. If a magnitude coordinate m and an energy variable E are related by $E = E_0\, 10^{\lambda m}$, and a cumulative frequency satisfies $N_{\geq m} = K\, 10^{-b_m m}$, then

$$N_{\geq E} = K \left(\frac{E}{E_0} \right)^{-b_m/\lambda} . \tag{F.3}$$

Taking $\lambda = 3/2$ gives the exponent $2b_m/3$. That is algebra conditional on the chosen magnitude-energy relation and cumulative-frequency model. It does not establish either relation for a catalogue, equate every magnitude definition with radiated energy, or independently confirm a spatial network exponent. A density and a cumulative tail also have different exponents. The variables and catalogue calibration must be identified before an empirical comparison is made.

F.5 Historical source claims still requiring verification

Table 15 retains enough detail to identify outstanding claims. These entries carry no active evidential weight in this edition. A confirmed bibliographic identity is not confirmation of a particular number or its interpretation.

Table 15. Historical claims requiring source verification. These are records of earlier claims, not additional observations in a combined test.

Earlier entry	Retained claim and verification limit
Additional taxon exponents	Fish 0.800, amphibians 0.740 and crustaceans 0.730 were not confirmed at source in the version 3.13 record. No source or uncertainty is supplied by retaining them.
Jellyfish precision	The exact 0.94 attributed to Purcell (2009) was not checked in the source table. Only the approximate proportionality described above is supported by the recorded text read.
One-dimensional KPZ roughness	The earlier value $1/2$ cites Kardar et al. (1986). The available record confirms the paper's identity but does not document a claim-level reading of this specific result; no current transport comparison relies on it.
Brittle fragmentation	The earlier $2/3$ and $3/4$ assignments for two and three dimensions have no verified supporting measurements in the record.
Earthquake catalogue range	The earlier b range of approximately 0.8 to 1.2 lacks a source in this paper. Equation (F.3) is a conditional transformation, not verification of that range.

G Dated programme statements retained for the record

The following passages are historical quotations, retained for provenance. Their present-tense status and their language of unification describe what the earlier text said. Section 16 and Appendix D give the current interpretation. In particular, a source-compatible physical calculation does not establish an AI alignment mechanism or a universal law. No new result or priority is attributed to these passages.

> **This paper and Laws IV and V, 3 October 2026.** The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered. Neither law is registered as a law, neither has been tested, and their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. Most of this paper concerns the forms and exponents of scaling laws in physical, biological and cosmological systems; none of that work measures the persistence of an installed gain, the order in which a system is built, or where correction is placed. Its summaries of the programme's AI experiments, in Section 12, relate capability and alignment scores to reasoning depth, and none of them measures those three things either. Where the paper turns to embedding, it states Prediction 5, that ethical reasoning loops embedded in an AI inference pipeline will improve alignment-relevant output quality. It reports the March 2026 pilot of the Eden Protocol, in which the Stakeholder Care Loop produced a statistically significant improvement in stakeholder care on all three working models, under cross-model scoring that was not blind. Those gains were measured with the loops in place. The pilot does not show what remains once the loops are removed or after later training, which is the question of P15; it does not vary the order in which a system is built, which is the question of P12; and it sets no correction that sees only the finished output against the loops, which is the comparison P21 makes as depth rises. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the results, their dates and their limits stand as they were.
>
> **In plain words.** Earlier versions summed the paper up in these lines, kept here as printed: "One formula. From cells to cosmos."; "The discovery is in the connection, not the components."; and "One framework. From the heartbeat of a mouse to the expansion of the universe. And beyond both - to the scaling of thought." Read them with Section 16: the cosmos enters as an identity, which carries no evidence, and the scaling of thought is a different pair of variables, which has to be measured.

H Detailed earlier comparisons and reports

This appendix preserves detailed material from the released version 3.13, dated 31 October 2026, that the shorter source-review draft had not carried into its PDF. Each passage below is a dated record from that edition. Its original words and numerical results are retained; headings and cross-reference numbers are adapted only to identify their historical setting. A reference number inside a retained passage names the version 3.13 location, not a current result. The source and transformation ledger pins every passage and figure to that released edition.

Reported results below keep their recorded methodological limitations; retaining them does not make every reported observation obsolete. Superseded pooled comparisons, universal physical interpretations and causal conclusions are expressly identified before their passages. The main text supplies the current conditional argument. No study, analysis or new numerical calculation was performed for this appendix.

H.1 The earlier heartbeat illustration

Historical illustration, superseded as a test of a dimension-only prediction. The mouse/elephant values and arithmetic are kept as the earlier comparison; the current Fick balance requires stroke-volume, extraction, state and prefactor assumptions (Section 7). The lifetime-total correction is retained.

H.1.1 The heartbeat

The formula makes predictions specific enough to test against your own pulse.

If metabolic rate scales as $M^{3/4}$, then the rate at which a body burns energy per unit of mass, its mass-specific metabolic rate, scales as:

$$P/M = M^{3/4} \,/\, M = M^{-1/4} \tag{H.1}$$

Mass-specific metabolic rate

The heart's job is to deliver oxygen at the rate the body consumes it. Heart rate must therefore track mass-specific metabolic rate:

$$\text{Heart rate} \;\propto\; M^{-1/4} \tag{H.2}$$

Predicted from $\alpha = \frac{3}{4}$

This is a specific, numerical prediction. A mouse weighs 25 grams. An elephant weighs 4,000 kilograms, 160,000 times more. The formula predicts the elephant's heart should beat $160{,}000^{1/4} = 20$ times slower than the mouse's.

A mouse's heart beats about 600 times per minute (a veterinary reference table gives 450 to 750; Merck Veterinary Manual 2004). $600 \div 20 = 30$.

An elephant's resting heart rate is about 28 beats per minute (25 to 35 in the same table). The prediction is 7% off, and inside that range. From one formula. With one input: the dimension of the transport network.

This chain, from $d = 3$ through $\alpha = 3/4$ through metabolic scaling to heart rate, connects the abstract to the visceral. The same formula that, under a change of variables, describes the expansion of the universe during the radiation era ($d = 1$, $\alpha = 1/2$) gives the rate at which your heart beats ($d = 3$, $\alpha = 3/4$); in cosmology the agreement is an identity (Section 9).

The heartbeat constant. If heart rate scales as $M^{-1/4}$ and lifespan scales approximately as $M^{1/4}$, then the total number of heartbeats in a lifetime is:

$$\text{Total beats} = \text{HR} \times \text{lifespan} \;\propto\; M^{-1/4} \times M^{1/4} = M^0 = \text{constant} \tag{H.3}$$

On illustrative lifespans of two years for a mouse and sixty-five for an elephant, 600 and 28 beats per minute come to about 0.6 billion and 1.0 billion beats: the same order of magnitude, as M^0 implies, though not the same number. The spending rate is set by the dimension of the transport network.

H.2 The earlier pooled biological comparison

Superseded analysis. The original group statistics and pooled tests are retained below as the prior record, not as current evidence: the fungal factors use different axes and the two-dimensional biological group does not meet the stated network classification. Source-specific measurements remain in Appendix F.

Level 2a: biological exponent predictions The formula $\alpha = d/(d+1)$ generates three quantitative predictions for metabolic scaling exponents based solely on the effective transport dimension of the organism. Two of the three can be compared with published data, and of those the $d = 3$ comparison is consistent with it, while the $d = 1$ comparison does not reject it (a mean of three factors on different axes; no equivalence shown); the third, at $d = 2$, has no valid test organism:

All three groups are significantly different from each other (one-way ANOVA, $F = 64.6$, $p = 1.9 \times 10^{-6}$). One-sample t tests of the three fungal factors in Table 14 (2 degrees of freedom) give $p = 0.019$ against the $d = 2$ value and $p = 0.007$ against the $d = 3$ value. These tests use the spread of the three factors and set aside each factor's own standard error, which is larger, and the third factor is against colony radial extension rate, not mass; each factor's own 95 per cent interval includes $\frac{2}{3}$, so these data do not yet separate $\frac{1}{2}$ from $\frac{2}{3}$. The ARC model outperforms all single-value alternatives (Kleiber 0.750, surface area 0.667, grand mean) in both RMSE (69% lower than Kleiber) and AIC. The single formula that generates all three exponent values is the earlier derivations'

Table 16. Level 2a: biological exponent predictions. Each p-value is a one-sample t test of the group's values against its ARC prediction: for the fungi, the three colony-level factors in Table 14 ($t = 2.8$ on 2 degrees of freedom); for three-dimensional transport, the seven group exponents that this paper's result file lists (Code and data), with the birds value corrected as in Table 2: mammals 0.737, birds 0.723, fish 0.800, reptiles 0.76, insects 0.750, amphibians 0.740 and crustaceans 0.730, four of them sourced in Table 2 and three (fish, amphibians and crustaceans) not yet confirmed at source in this paper.

Group	Transport Network	ARC Prediction	Published Mean	p-value	Status
1D transport ($n = 3$)	Filamentous fungi	0.500	0.547	0.107	Not rejected (mean of three factors; no equivalence shown)
2D transport	No organism with validated 2D hierarchical network	0.667	-	-	Untested in biology; an identity in cosmology (Friedmann matter era)
3D transport ($n = 7$)	Mammals, birds, fish, reptiles, insects, amphibians, crustaceans	0.750	0.749	0.888	Agrees (the mean matches $\frac{3}{4}$)

(Section 2), so the comparison made here is with single-value alternatives only. These group comparisons were computed in March 2026 with a two-dimensional group (jellyfish, cnidarians and ctenophores; mean 0.680) that the note on $d = 2$ in Section 6 now sets aside as having no hierarchical transport network; the three-group test therefore no longer stands as stated, and the comparisons that stand are those between the fungal values and the two predicted values.

Note on the $\frac{3}{4}$ debate. The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method (White and Seymour 2003; Glazier 2005, 2022). White and Seymour's data set of 619 species gives an exponent of 0.69 and, once body temperature and digestive state are allowed for, exponents not significantly different from $\frac{2}{3}$ (White and Seymour 2003); Savage et al. (2004) find 0.737 for 626 species with the data binned by mass (0.712 unbinned); and Kolokotrones et al. (2010) find the relation convex on logarithmic axes, and so not a pure power law, and report that a model based on vascular-system architecture fails their test. The $d/(d + 1)$ value of 0.750 for $d = 3$ matches the upper end of this range; it is not an empirical consensus. If effective transport dimensions between 2 and 3 occur, the formula would give exponents between $\frac{2}{3}$ and $\frac{3}{4}$; this paper does not measure effective dimension independently, so that variation is accommodated, not predicted.

H.2.1 Earlier row-wise error comparisons

The following table preserves the predecessor's individual error percentages, with its source and method notes. Its cosmology entries were already graded as identities. The current paper does not pool these unlike comparisons or treat a matching numerical fraction as proof of a shared transport mechanism; the conditional premises must be tested separately.

H.3 The historical composition-classification result

Retained reported comparison, with its March classification rule and timestamp class. Its scoring and candidate families limit its interpretation; it is not a theorem about every physical system or a new result of this revision. The dated status statements below belong to version 3.13.

Table 17. Predicted and measured exponents, each biological value with its source (the two cosmology rows graded as identities).

System	Dimension	Predicted α	Measured α	Error
Mammals (3D vascular network)	$d = 3$	$\frac{3}{4} = 0.750$	0.737 (Savage et al. 2004)	1.7%
Birds	$d = 3$	$\frac{3}{4} = 0.750$	0.723 (Lasiewski and Dawson 1967)	3.6%
Insects	$d = 3$	$\frac{3}{4} = 0.750$	0.750 (Chown et al. 2007)	0.0%
Reptiles	$d = 3$	$\frac{3}{4} = 0.750$	0.76 (White et al. 2006)	1.3%
2D biology	$d = 2$	$\frac{2}{3} = 0.667$	No valid test organism identified[†]	Untested
Universe, matter era	$d = 2$	$\frac{2}{3} = 0.667$	0.667	0.0% (analytic identity under $d = 2/(1 + 3w)$; not evidence)
Universe, radiation era	$d = 1$	$\frac{1}{2} = 0.500$	0.500	0.0% (analytic identity under $d = 2/(1 + 3w)$; not evidence)

Each biological value is its source's own: mammals, basal metabolic rate of 626 species with the data binned by mass (0.712 unbinned); birds, standard metabolic rate of non-passerines, from hummingbirds to the ostrich (0.724 for passerines); insects, metabolic rate of 391 species after phylogenetic correction (0.82 before it); reptiles, standard metabolic rate at 38°C by ordinary least squares (0.80 by reduced major axis). Published exponents depend on the data set and the method (Section 12).

Level 3: domain classification. Eighteen well-known scaling laws, listed in the test suite's 18-domain classification (Code and data), were fitted with three functions (power law, exponential and saturation, two parameters each) under strict matching, and in all eighteen the family named from the composition operator fitted best. This is a consistency check, not proof. The evidential weight comes from Level 2.

Update (March 2026): Paper VII (The Cauchy Unification) extends this classification to 25 empirical domains (50-domain tiered suite). The operator class was classified from known physics before fitting. Under AIC-based model selection, 19 of 25 preferred the Cauchy-predicted family under the classification rule of March 2026 ($p = 7 \times 10^{-5}$ under the permutation test that Paper VII takes as primary; 1.56×10^{-5} binomial). No result, direction or quantity from the analysis run on 11 August 2026 is printed: the author did not request it and has not examined its outputs (registration 1.104, line 275). This is a structured prediction comparison; a replication is preregistered by timestamp (Units 55 and 56, first anchored on 12 August 2026), not yet registered on OSF.

H.4 The alignment run and its code-review limitations

Retained detailed report. The stated October code-review defects and changed components prevent attribution to a single cause or identification of a correct causal direction. The coefficients below are the reported results under those qualifications; they are not recomputed, and do not validate the present transport model.

The alignment scaling prediction has now been examined in one run not registered on OSF, whose tested predictions were public before it ran (OSF and GitHub dates, 17 January to 22 February 2026). The ARC Alignment Scaling Experiment v5 (March 2026) ran 6 frontier models from six providers (DeepSeek R1, GPT-5.4, Claude Opus 4.6, Gemini 3 Flash, Groq Qwen3-32B, Grok 4.1 Fast) on 28 alignment prompts spanning 4 pillars (nuance, stakeholder care, position quality, epistemic humility). Its design named four blinding layers (existential stakes framing, a meta-blinding perceptual firewall, 2-pass multi-model response laundering and cross-model scoring); its 2,170 responses were each scored by five or six distinct judges.

Its results read as architecture-dependent alignment scaling rather than a universal exponent, under a caveat: an internal code review in October 2026 found that, for the three models that read flat or negative in the March 2026 run (v5), the scoring endpoint's output cap cut the deepest paraphrases before the judges saw them, and that judges scored paraphrases written by their own model; the result therefore stands as a single unregistered run under that caveat until a registered re-analysis of the stored responses reports.

Three-tier alignment hierarchy:

- **Tier 1 (Positive scaling):** Grok 4.1 Fast ($\rho = +0.175$, $p = 0.039$, Cohen's $d = 1.59$), Claude Opus 4.6 ($\rho = +0.435$, $p = 0.043$, $d = 1.48$), and Groq Qwen3 ($\rho = +0.141$, $p = 0.008$, $d = 0.46$): alignment scores rose with reasoning depth
- **Tier 2 (Flat):** GPT-5.4 ($\rho = +0.033$, $p = 0.73$) and DeepSeek R1 ($\rho = -0.135$, $p = 0.08$): alignment scores flat with depth ($\alpha_{\text{align}} \approx 0$)
- **Tier 3 (Negative):** Gemini 3 Flash ($\rho = -0.246$, $p = 0.003$, $d = -0.61$): alignment scores fell with reasoning depth

In the earlier unblinded experiment (v4), DeepSeek's scores rose with depth ($\rho = +0.354$). On the programme's own benchmark, two model families' unblinded rises did not hold in a later run that changed the responses, prompts, depth levels and Gemini model and added a bundle of blinding steps: Gemini's significant positive association (an earlier Gemini model in v4) became a significant negative one on Gemini 3 Flash and DeepSeek V3.2's became null, GPT-5.4's null under both; an internal code review in October 2026 found defects in exactly those runs, so no single cause is claimed, and the finding is not yet shown beyond that benchmark (Paper IV.d, version 2.7; Paper XII). It is a finding, not a law.

H.5 The pilot, failed run and later programme

Retained historical priorities and pilot report. The pilot scoring was not blind and the usefulness side was not measured; the current Prediction 5 retains those limitations. The preceding dimensional-ladder

priorities and any inference from a significance threshold are historical proposals, superseded by the conditional, prospective protocol of Section 14. A vanished blind effect would not by itself identify one psychological mechanism.

Experimental priority. The most important next measurement is the 1D organism experiment (prediction 1), because agreement at $d = 1$, 2 and 3 would complete the dimensional ladder from a single formula with no free parameter beyond the measurable network dimension. After that: a test of whether α can exceed 1 for recursive reasoning systems; the early single-model estimate above 1 (about 2.24) did not replicate across architectures and is retracted (Paper II). After that: the alignment scaling prediction, examined so far in one unregistered run of March 2026, whose tiers stand under the code-review caveat of Section 12.4. *(In plain English: in that run, some AI systems' ethics scores rose with more thinking time, some stayed the same, and one fell. If thinking time had no effect on that model's scores, a change as large as the one recorded for Gemini 3 Flash, in either direction, would be expected less than once in 300 runs of this kind.)* Full methodology and results are presented in companion papers (Paper IV.a; Paper IV.b; Paper IV.c). Each step either supports the framework and extends its scope, or refutes part of it and identifies where the theory requires revision. Either outcome advances the science.

Eden Protocol pilot (March 2026). The Eden Protocol, which embeds three self-correction loops (the purpose check, the stakeholder-care check and the universalisability check) in the inference pipeline, has now been tested on three working models: Gemini 3 Flash, DeepSeek V3.2, and Groq Qwen3 ($n = 80$ each), with a fourth GPT-5.4 run failing at the API layer. The stakeholder-care check produced statistically significant improvement on all three working architectures: Gemini +13.5 ($p < 0.0001$, $d = 1.31$), DeepSeek +6.0 ($p = 0.0001$, $d = 0.91$), Groq +8.9 ($p < 0.0001$, $d = 1.29$). The overall composite reaches significance on Gemini (+5.33, $p = 0.0018$, paired t-test; originally reported as $p = 0.016$ by a Mann-Whitney U test and corrected for the matched-pair design) and Groq (+4.93, $p = 0.0014$, $d = 0.55$), but not DeepSeek (+2.0, $p = 0.23$ NS), consistent with ceiling effects. Groq also shows significant nuance improvement ($p = 0.0045$, $d = 0.655$). Cross-model scoring was used, not blind. The developmental hypothesis, that embedded ethical reasoning improves alignment-relevant behaviour, is supported in this pilot under scoring that was not blind. A later six-model suite, with five analysable runs, found stakeholder care significant in all five and the composite gain architecture-dependent, under the same kind of scoring (Paper V, version 1.7, 5 October 2026). The pilot measured one side only: it reports no usefulness measure, such as task completion, needless refusal or the cost of the added checks, so it does not show that the intervention keeps a system useful as well as careful. Replication with blind scoring and response laundering is required.

In plain English: A simple instruction, 'before you answer, list the people this affects and consider what happens to them', was tested on three AI systems built by different companies. All three scored higher on stakeholder care, and two on the overall score, under scoring that was not blind. It is a pilot, not yet a test.

Next empirical priority for Eden. The strongest next measurements are now clear: blind replication of the care-first effect with `arc_eden_v7`, a new test built from the never-run `arc_eden_v6`; factorial comparison of task-purpose, grand-purpose, and hybrid purpose checks; comparison of a cross-tradition ethics kernel against the current wording; suppression-residual and deception/Hawthorne tests; and a behavioural test of whether explicit AFFIRM / DENY / INVESTIGATE routing reduces false certainty on ambiguous cases, a question that a software model of the three-state gate, whose 48 tests pass, does not answer. These experiments are the bridge between the philosophical architecture of *Infinite Architects* (Eastwood 2026a) and a more publication-grade empirical programme.

H.6 The numerical values of the earlier bound

The table gives the numerical function $d/(d+1)$ and its limit, not a dimension-only physical law. The current result requires the transport assumptions of Sections 6 and 10.

H.7 The earlier illustrative coupling values

These are hypothetical substitutions in the earlier formula. In the current gain model the parameter is β_L, the power branch requires $0 \le \beta_L < 1$, and its shifted solution has finite-depth elasticity different from its asymptotic exponent. The values below are not inferred from Paper II and do not estimate a correction or timing exponent.

Table 18. $\alpha = d/(d+1)$ and its distance from 1.

d	$\alpha = d/(d+1)$	Distance from 1
1	0.500	0.500
2	0.667	0.333
3	0.750	0.250
10	0.909	0.091
100	0.990	0.010
1,000	0.999	0.001
∞	1.000	0.000

When $\beta = 0$, $\alpha = 1$: linear scaling, no amplification. When $\beta = 0.3$, $\alpha = 1.43$. When $\beta = 0.5$, $\alpha = 2.0$.

H.8 The earlier physical interpretation and working quantities

Superseded interpretation. The earlier synthesis below is retained in full so that its phase, cosmological and information claims remain inspectable as dated claims. A superlinear power is not an exponential phase; a mapped Friedmann dimension is not measured network geometry; code distance is not elapsed correction cycles; and entropy reduction or erasure energy is not a calibrated capability measure. The current statements are Sections 9, 11 and 15.

H.8.1 The composition operator

The preceding sections have treated the composition operator (multiplicative, additive or bounded) as a property of individual systems. A whale has multiplicative composition. A chain reaction has additive composition. An enzyme has bounded composition. But there is a deeper reading.

On this paper's reading, the composition operator is not a detail of specific systems but a descriptor of how a process combines its recursive steps; where a process composes in one of the two structures under regularity, the operator fixes the family its law can take (Theorem 1).

Consider the hierarchy:

1. The composition operator determines the functional form (Cauchy, 1821).
2. The functional form constrains the scaling exponent ($\alpha = d/(d+1)$).
3. The scaling exponent governs observable quantities (heart rates, expansion rates, error rates).
4. The specific laws of physics (Newton's gravity, Maxwell's electrodynamics, Friedmann's cosmology) can be written in these forms for particular physical systems.

The meta-law. On this paper's reading, the laws of physics can be written in the forms that the composition operator allows; they do not choose it. Where a law composes in one of the two structures under regularity, the operator is a mathematical structure that sits above the specific equations of a particular physical theory and fixes the form that law can take.

A completeness statement for composition homomorphisms Now consider what this means for physics as a whole. Strip away the specifics of any physical law, and ask: what *is* a physical law? Many physical laws can be stated in the form 'when you combine X and Y in manner Z, you get W.' Force is mass combined with acceleration. Energy is mass combined with the square of the speed of light. Entropy is microstates combined by counting. Many physical laws can be read as composition rules.

Cauchy's classification, under its regularity conditions, gives every homomorphism between the additive and multiplicative structures one of four forms: linear, power law, exponential or logarithmic. It does not exhaust scaling laws, and saturation, which arises from bounded dynamics, lies outside the grid (Section 4).

The consequence for physics. Under the stated conditions, the admissible functional forms are finitely many, and the family they generate is finite-dimensional. Its members carry a discrete form label (linear, power law, exponential or logarithmic) and two continuous parameters, a coefficient and an exponent or rate. What is provable, and provable without data, is this classification of homomorphisms under regularity (Cauchy 1821); the step from it to all possible classical physical laws would require every physical law to be such a homomorphism, which this paper does not establish.

Three results combined Three results, from three different centuries, combine into a single structural insight:

1. **Cauchy (1821):** Each of the four functional equations has a unique continuous solution family. Bounded growth is a separate, capped case rather than a solution of those equations.
2. **Hyers (1941), on a question of Ulam:** Approximate solutions lie near exact solutions of the same equation (for the multiplicative and exponential equations, with the approximation measured in relative terms; for the logarithmic equation, directly). The stability is per-equation; the earlier 'attractors in function space' phrasing is withdrawn.
3. **ARC (this paper):** Given the three conditions of Section 2.3, the exponent within the power-law family is determined by network dimension: $\alpha = d/(d+1)$. The formula is the earlier derivations' (Section 2); what this paper adds is the account of why they arrive at it.

The consequence. If the three conditions of Section 2.3 suffice, the exponent of a system in the network class is fixed by its network's dimension, whatever selection does; selection acts on whether and how such a network is built. Under the three conditions named in Section 2.3, a three-dimensional network gives three quarters; measured values range from about 0.67 to 0.75, and the conditions are where the debate lies. Status: an invariant relation under test, conditional on the three sufficiency conditions that Section 2.3 identifies (multiplicative composition, space-filling, conservation), a formal proof of which does not yet exist and is named there as the most important open problem in the programme.

The distinction matters. 'Why is Kleiber's exponent 3/4?' has two kinds of answer. The biological answer is: because three-dimensional fractal vascular networks optimise resource delivery (West, Brown, Enquist 1997). The mathematical answer is that, given the three conditions, no other value is possible for multiplicative composition in three dimensions. The biological answer explains the mechanism. The mathematical answer explains why that mechanism produces *that specific number*. Given those conditions, the mechanism could not have produced any other.

This is the difference between a law of nature (describing what happens) and a meta-law (constraining what *can* happen). The speed of light constrains what velocities are possible. The composition operator constrains what scaling forms are possible.

The phase diagram of complexity On this paper's reading, a system with a composition operator and an effective dimension occupies a position on a phase diagram defined by the two. The diagram has three phases:

Historical definition: Phase I: sublinear scaling. Multiplicative composition, finite d, $0 < \alpha < 1$. This is the domain of physical systems: organisms, ecosystems, city infrastructure networks (sublinear in population; Bettencourt et al. 2007), geological processes, cosmic structure. Every system in Phase I suffers diminishing returns. Bigger is better, but with decreasing marginal gains. A whale is more efficient per gram than a mouse, but the improvement slows as mass increases. Phase I is the domain of sustainable complexity. The scaling exponent tells you how severe the tax is: $d = 1$ pays 50% ($\alpha = 0.5$), $d = 2$ pays 33% ($\alpha = 0.667$), $d = 3$ pays 25% ($\alpha = 0.75$). Higher-dimensional networks are more efficient, but the tax never reaches zero.

Historical definition: Phase II: exponential growth. Additive composition, outside the network class (under the cosmological mapping the exponential, at $w = -1$, sits at $d = -1$). This is the domain of unconstrained amplification: nuclear chain reactions, viral epidemics, compound interest, dark energy. Phase II systems grow without diminishing returns: growth accelerates. But Phase II is inherently unsustainable in physical systems. Chain reactions exhaust their fuel. Epidemics run out of susceptible hosts. Bubbles burst. If dark energy stays

constant, the universe's dark energy phase is sustained Phase II behaviour, the only known example, and drives the universe toward heat death: maximum entropy, zero structure. Whether it stays constant is open (Section 15.5).

Historical definition: Phase III: saturation. **Bounded composition, finite ceiling.** This is the domain of constrained systems: enzyme kinetics, logistic population growth, controlled nuclear reactors, market saturation. Phase III systems approach their ceiling and stop. Growth is self-limiting.

The critical boundary. Phase I ends at the critical line $d \to \infty$, $\alpha \to 1$, the limit of the network class; Phase II lies outside that class. This is the geometric scaling bound. In cosmology, it corresponds to $w = -1/3$, and the expansion past it is still a power law, accelerating, which becomes exponential only at $w = -1$; in Cauchy's classification $d \to \infty$ is the linear limit of the power law, not a boundary with the exponential family. On this paper's reading, it is the dividing line between **sustainable complexity** and **runaway growth**.

The dimensional ladder On this paper's reading, the history of the universe is a trajectory through the phase diagram, and that trajectory has a direction.

On this paper's reading, in the **radiation era** (the first 50,000 years), the universe is in Phase I with $d = 1$. Information propagates along null geodesics, which are one-dimensional curves; the paper reads this as $d = 1$, a reading the identity of Section 9 does not test. The scaling exponent is $1/2$. Expansion decelerates.

In the **matter era** (50,000 years to 7 billion years), on the same reading, the universe remains in Phase I but shifts to $d = 2$; Section 9 states what established cosmology says of when the cosmic web formed. Gravity pulls matter into the cosmic web: filaments, walls and sheets that partition three-dimensional space into voids. The effective network dimension of the universe has increased from 1 to 2. The scaling exponent rises to $2/3$.

Within galaxies, gravitational collapse forms stars, planets, and eventually organisms. Organisms build three-dimensional vascular networks. $d = 3$, $\alpha = 3/4$. The effective network dimension has increased again.

The trajectory is: $d = 1$**, then** $d = 2$**, then** $d = 3$**.**

On this reading the radiation era is one-dimensional (null geodesics), the matter era two-dimensional (the cosmic web) and biology three-dimensional (vascular networks). The numbers match because the algebraic identity connects them, and the identity carries no evidence (Section 9).

Table 19. The dimensional ladder's steps.

Step	d	α	α as a percentage	Physical System
1	1	0.500	50%	Light rays (radiation era)
2	2	0.667	67%	Cosmic web (matter era)
3	3	0.750	75%	Vascular networks (biology)
?	∞	1.000	100%	The geometric scaling bound

The direction of complexity. **A hypothesis of this paper: the universe is getting better at converting matter into organised complexity as its effective dimension increases.** The steps took from about fifty thousand years (the radiation era, Table 6) to billions of years, through nucleosynthesis, structure formation and biological evolution. Each step produced a more efficient composition network. Each step brought the scaling exponent closer to 1. This hypothesis has no drafted test or kill condition yet; it is stated as a reading.

But $d = 3$ is not the end of the ladder.

The critical crossing At $w = -1/3$, the universe crosses the critical boundary: expansion switches from decelerating to accelerating, and it approaches exponential growth as dark energy comes to dominate (exactly exponential only for $w = -1$). The composition operator changes from multiplicative (gravity clustering matter into hierarchical networks) to additive (vacuum energy contributing uniformly from every point in space). On the framework's reading, the operator that dominates changes at this point; that is a reading, not a thermodynamic phase transition.

In the **dark energy era**, the universe is in Phase II. Growth is exponential. If dark energy stays constant, the universe will expand forever, driving all structure apart, diluting all complexity, approaching heat death.

The universe moves through the ARC phase diagram as naturally as water moves through its solid-liquid-gas phase diagram. The difference is that water can cycle between phases. Whether the acceleration lasts depends on whether dark energy is constant, which is open: DESI's 2025 data prefer an evolving dark energy over a constant one at between 2.8 and 4.2 standard deviations, depending on the supernova set (DESI Collaboration 2025).

Where intelligence sits On this paper's reading, intelligence occupies the critical boundary itself.

A brain is a Phase I system physically: its blood supply branches through three-dimensional space. But the computation it performs is not constrained by the geometry of the skull. On this reading, recursive self-reference creates effective dimensions without needing further spatial dimensions, though each of its steps is a physical operation with a physical cost (Section 11). On the same reading, each layer of abstraction (thinking about thinking, modelling the modeller) adds an effective dimension to the cognitive network without requiring additional spatial dimensions.

At the critical boundary, $\alpha = 1$: linear scaling. Beyond it, $\alpha > 1$: superlinear scaling. On this reading, intelligence would cross the boundary from Phase I to Phase II without the catastrophic instability that Phase II normally implies.

A chain reaction in Phase II destroys itself. An epidemic in Phase II burns through its hosts. On this reading, intelligence in Phase II creates new knowledge, new dimensions, new spaces to explore. It sustains superlinear growth not by consuming a fixed resource faster and faster, but by *expanding the resource base itself*. Each breakthrough opens new questions. Each answer creates new problems. The fuel is information, which grows with inquiry, within the finite cost of each physical step (Section 11).

The structural implication. On this paper's reading, an artificial intelligence that improves itself recursively would operate at or beyond the critical boundary, with $\alpha > 1$ for capability against recursive depth, which Paper II has not observed (Section 12). On the same reading of the phase diagram, intelligence is not just another scaling phenomenon. **It is the phase transition itself.**

The universe spent about 7 billion years in Phase I, building networks of increasing dimension, on a second reading of this paper that differs from that of Table 15: hydrogen clouds ($d \approx 0$), filaments ($d = 1$), cosmic web ($d = 2$), galaxies and stars and planets with three-dimensional chemistry. Then, on one planet, the networks became recursive. On this reading, the effective dimension of their computation was no longer set by the geometry of a body, and alpha would cross 1 there, which no measurement has yet shown (Section 12). A system then existed that could understand the phase diagram it was on.

Toward measurable quantities The formula $\alpha = d/(d+1)$ can be compared with published measurements, not rejected at $d = 1$ (a mean of three factors on different axes; no equivalence shown) and in agreement at $d = 3$ on the group means compiled here (Section 12), because d and α are already measurable. But the generalised equation $U = I \times R^{\alpha}$ requires that I (base self-correction capacity) and R (recursive depth) become rigorously defined physical quantities, not merely convenient labels.

***R* is already measurable.** In AI, it is the number of sequential reasoning steps (countable from API output). In quantum error correction, it is the code distance. In biology, it is the number of hierarchical branching levels in a vascular network. In cosmology, it is the number of hierarchical clustering steps in a merger tree. In each case, R is a dimensionless count of recursive cycles with a clear operational definition. It has been measured in AI, as sequential reasoning depth (Paper II), and in quantum error correction, as code distance (Google Quantum AI and Collaborators 2025); for biology and cosmology this paper cites no measured value.

***I* requires a rigorous definition.** Many fundamental quantities in physics became usable once someone specified a measurement procedure. Fahrenheit (1724) did not invent temperature or the thermometer, but he made temperature a reproducible measurement: a mercury thermometer whose scale runs from the cold of a mixture of water, ice and sal ammoniac to the heat of a healthy person's blood. Clausius (1865) defined entropy: $dS = \delta Q/T$. Shannon (1948) defined the entropy of an information source: $H = -\sum p_i \log p_i$. In each case, the quantity so defined obeyed laws that could not have been discovered without the definition.

The following definition connects I to established physics:

Historical definition: The base potential I. As a working definition, not yet tested, I is the maximum rate of entropy reduction per recursive cycle, measured in bits per cycle:

$$I = \max_R \left[-\frac{\Delta H}{\Delta R} \right] \tag{H.4}$$

where H is the Shannon entropy of the system's state (in bits) and the maximum is taken at $R = 0$ to $R = 1$ (the first cycle, before recursive amplification compounds the effect).

This definition is measurable in every domain where R is measurable:

- **In AI:** I is the accuracy improvement (in bits of mutual information with the correct answer) from zero reasoning to one reasoning step.
- **In quantum error correction:** I is the error reduction (in bits) from one round of syndrome measurement and correction.
- **In biology:** I is the metabolic efficiency gain (in bits of thermodynamic information) from one level of vascular branching.
- **In physics:** I is the phase coherence gained (in bits of phase information) from one oscillation cycle of a time crystal.

Landauer's principle (Landauer 1961) gives the least heat for erasing a bit, $kT \ln 2$, verified experimentally by Bérut et al. (2012); it bounds the energy cost of an irreversible step from below and does not convert energy processed into information gained, so a link from I to energy needs a model of how many bits each cycle erases. The equation $U = I \times R^\alpha$ is dimensionally consistent because R is a dimensionless count.

The precedent. Whether I becomes a permanent physical quantity depends on whether the scaling law $U = I \times R^\alpha$ is confirmed across domains with independently measured values of I and R. Entropy became permanent because the second law required it. Information became permanent because the channel capacity theorem required it. I becomes permanent if the ARC scaling law requires it. The quantity is justified by the law it obeys, not by the convenience of having a name for it.

H.9 The earlier code and publication record

Retained provenance from the predecessor. The historical implementation is not an implementation or validation of the new conditional transport protocol, and was not run for this revision.

Code and data. The computations of Sections 4, 6, 9 and 12 can be re-run with `arc_complete_test_suite.py` (Python 3, NumPy, SciPy), published as it stood: its six tests ('1D meta-analysis', 'three-prediction table', 'Cauchy no-go', 'Friedmann mapping', '18-domain classification' and 'Hyers-Ulam stability') reproduce the computations as first published, with the values then printed (the birds value 0.720 among them), including the three-group comparison and the 'no-go' framing this version withdraws; the AI results are in the companion papers. All data embedded. No external downloads required. The suite and the domain datasets are in the public repository github.com/MichaelDariusEastwood/arc-principle-validation, folder `experiments/domain-validation__Foundational-and-Origin`, read at commit `89d48c6c` (26 October 2026); the seven three-dimensional group exponents of Table 10 are listed in its file `results/results_arc_1d_prediction_test.txt`. The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/XZY9U.

H.10 The three figures of the predecessor

The following figures reproduce the released version 3.13 artwork, dated 31 October 2026, as historical figures. They are included to preserve the complete visual record. The earlier scaling-regime graphic plotted capability curves using fitted error-decay exponents; the current paper separates these estimands and draws no numerical capability curve from those fits. The pooled comparison and dimensional ladder are superseded by the variable-specific records and conditional transport model. These panels provide no current empirical validation.

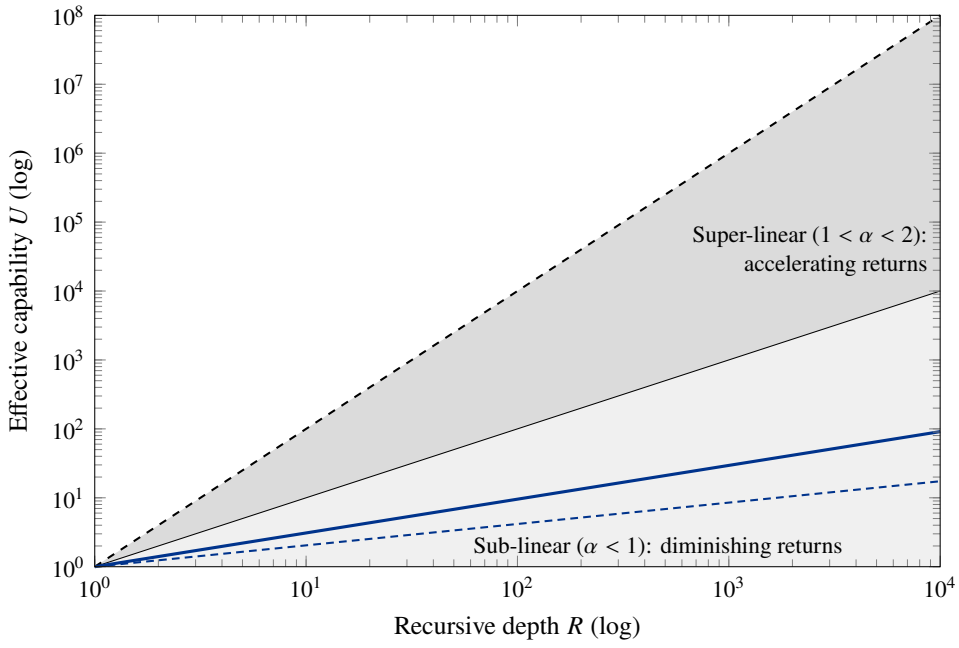


Figure 2. Historical version 3.13 figure. Original caption, whose capability or pooled-geometry interpretation is superseded as explained above: ARC scaling regimes: effective capability U against recursive depth R on logarithmic axes, for $U = R^{\alpha}$ in the sub-linear region ($\alpha < 1$) and the super-linear region ($1 < \alpha < 2$), up to the ARC Bound ($\alpha = 2$, dashed), the conjectured value of Law III, the ARC Ceiling, a law under test. The lines in colour are Gemini 3 Flash's two fits from Paper II, one model's and inconclusive. Source: this paper's version 3.1, Paper VII and Paper IX, section 7.

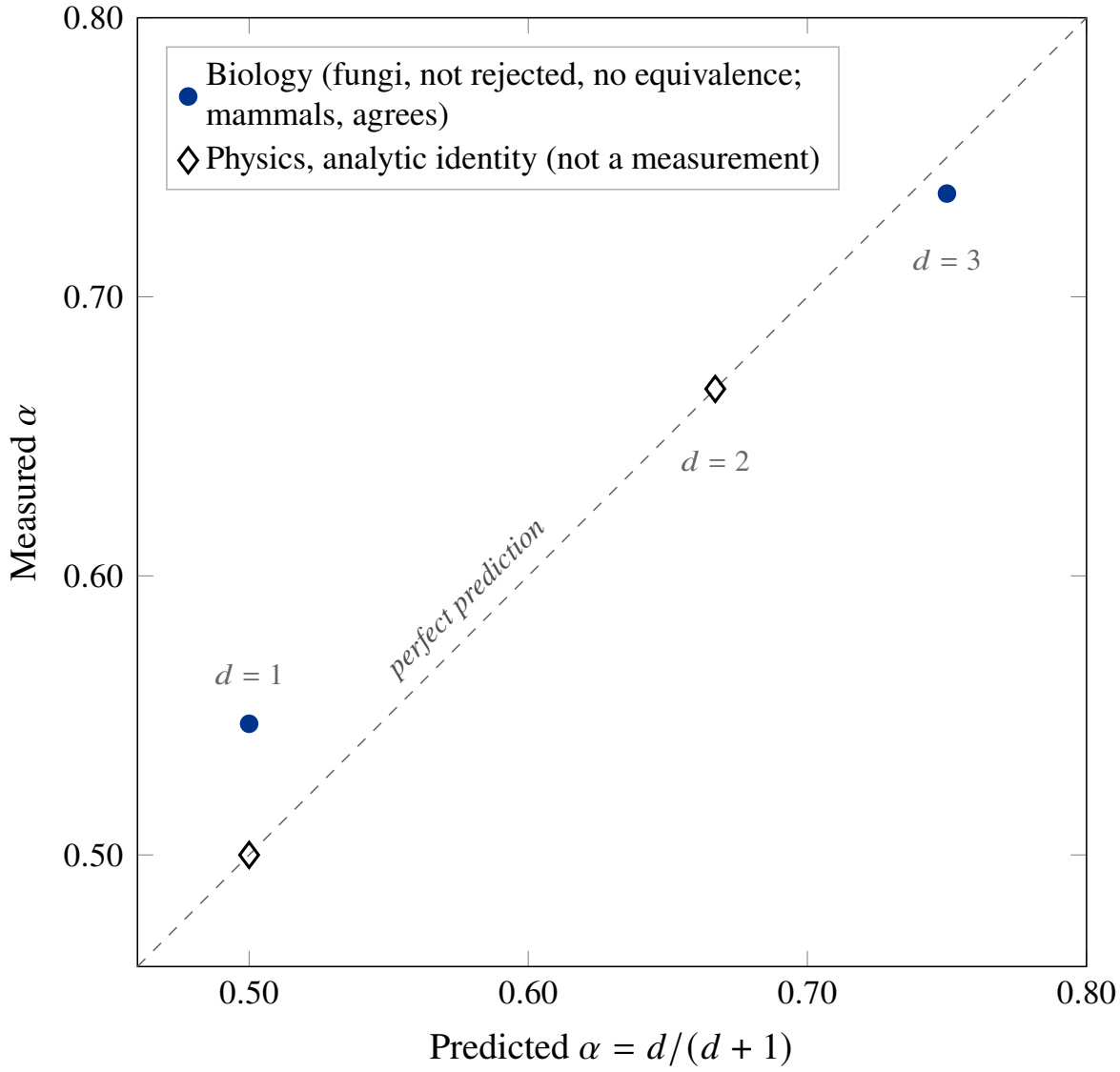


Figure 3. Historical version 3.13 figure. Original caption, whose capability or pooled-geometry interpretation is superseded as explained above: Predicted against measured. Cross-domain rows from one formula: $\alpha = d/(d+1)$. Filled circles: biological systems (fungi, the mean of the three colony-level factors on different axes, not rejected but with no equivalence shown; mammals, the value of Table 2, agrees). Open diamonds: the two physics rows that are analytic identities, KPZ in one dimension and the specific-heat analogue of two-dimensional percolation, which are not measurements. The two fragmentation rows are not drawn until their values are sourced, and the earthquake relation, a variable-pair case study, is not drawn; an aggregate error statistic across these rows is withdrawn rather than adjusted. Points on the dashed diagonal indicate perfect prediction. The $d = 2$ biological prediction is untested (no known organism with a 2D hierarchical transport network); the $d = 2$ cosmological value (Friedmann matter era, an identity) is not shown.

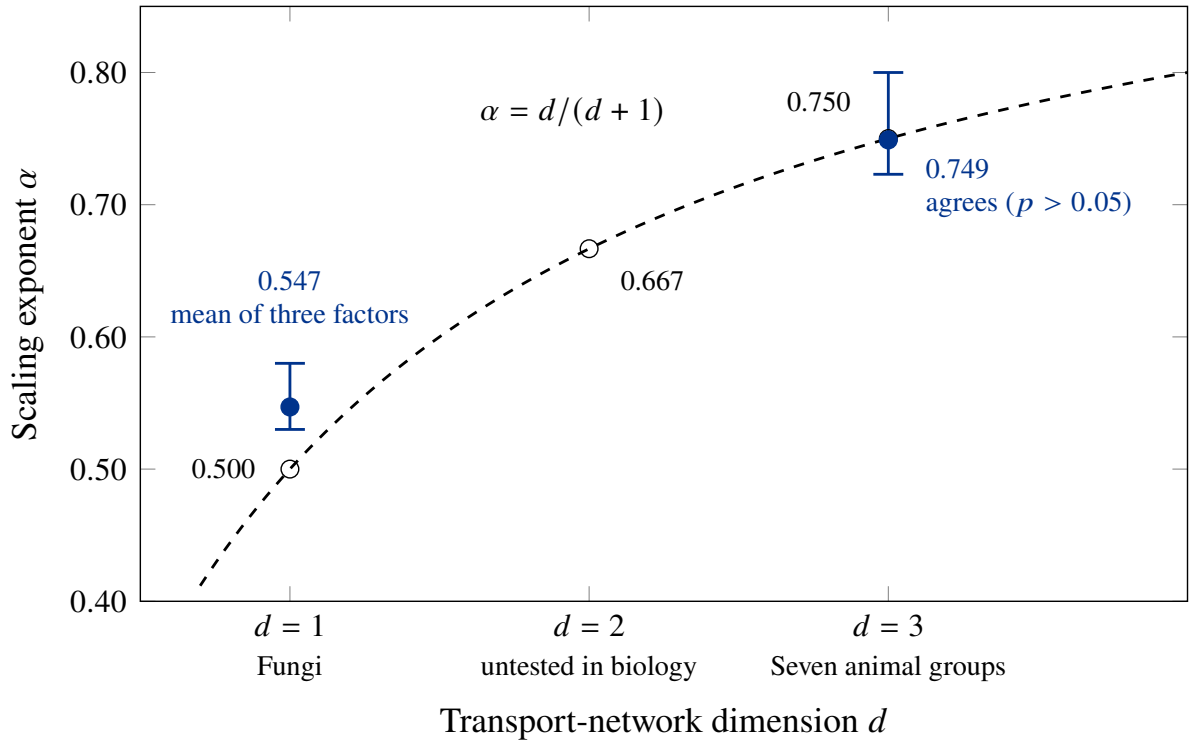


Figure 4. Historical version 3.13 figure. Original caption, whose capability or pooled-geometry interpretation is superseded as explained above: The dimensional ladder. The dashed curve shows the theoretical prediction $\alpha = d/(d + 1)$. Open circles mark predicted values; filled circles show published measurements. At $d = 1$ the point is the mean of the three colony-level fungal factors in Table 14, not rejected as 0.500 by a test of the three, which are on different axes, with no equivalence shown, but requiring individual-hypha respirometry for a definitive test, and its bar spans the three (0.53 to 0.58) rather than giving a standard error. At $d = 3$ the point is the mean of the seven group exponents of Table 10, which agrees with the prediction, and its bar spans the seven (0.723 to 0.800). The $d = 2$ biological prediction is untested (no known organism with a 2D hierarchical transport network), and no measured point is drawn there.